Estimating Local Average Treatment Effects Under Weak Monotonicity

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Abstract

I study the estimation of average treatment effects in the presence of compliers and defiers who occupy distinct covariate strata. I consider the reordered IV estimator suggested by Słoczyński (2026), which reorders the instrument for covariate values where defiers are present. I characterise the asymptotic properties of a feasible reordered IV estimator where propensity score estimates are used to identify defier covariate strata. I show conditions under which the feasible estimator is asymptotically equivalent to an oracle estimator with knowledge of defier covariate strata. I propose estimating the propensity score using flexible semiparametric methods, and show that the resulting estimator can achieve oracle equivalence. I study the finite sample properties of this estimator in a simulation exercise, and apply it to the data of Card (1993).

1 Introduction

Instrumental variables generate exogenous variation in a variable of interest while having no direct causal effect on the outcome. In the heterogeneous treatment effect model, variation arises as the instrument affects the treatment decisions of a group of switchers. Switchers may be compliers, who are induced to opt-in to treatment by the instrument, or defiers, who are induced to opt-out. When both types are present, the instrument induces flows in and out of treatment. If compliers cannot be distinguished from defiers, then variation in the instrument identifies some mixture of the effect of treatment take-up for compliers and treatment drop-out for defiers. A quantity that may not correspond to the treatment experienced by anyone in the population. In a setting without covariates, Imbens and Angrist (1994) resolves this by adopting the monotonicity assumption: only

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one type of switcher exists. The instrumental variable then identifies an average of the treatment effects experienced by that group, and this average can be estimated by two-stage least squares.

In the heterogeneous treatment effect model with covariates, the instrumental variable identifies an average treatment effect if the monotonicity assumption holds conditional on covariates. Both compliers and defiers may be present, but not in the same covariate strata. Słoczyński (2026) christens this 'weak monotonicity'. In this setting, the type of switcher is identified by the probability of treatment conditional on covariates and the instrument. That is, the propensity score. For a given covariate strata, if switchers are compliers, then the propensity score is increasing with the instrument. If they are defiers, it is decreasing. This allows an average treatment effect to be estimated for each group and, in turn, an overall average.

Słoczyński (2026) points out that, under weak monotonicity, the two-stage least squares specification commonly used in empirical work does not estimate a weighted average of treatment effects. Heckman and Vytlacil (2005) and Kolesár (2013) show that, for the two-stage least squares estimand to correspond to a weighted average of treatment effects, the first stage must accurately approximate the effect of the instrument on the propensity score insofar as the sign is correct: a change in the instrument which increases the propensity score must also increase the first stage prediction for treatment. The standard two-stage least squares specification omits covariate-instrument interactions. In doing so, it imposes that the sign of the effect of instrument on the propensity score is constant. With two types of switchers, this does not hold. As a result, one type of switcher receives negative weights in the two-stage least squares estimand.

In response to this Słoczyński (2026), advocates the saturate and control specification of Angrist and Imbens (1995), where covariates are converted to a single categorical variable that is interacted with the instrument. This specification essentially estimates a separate first stage regression for each covariate strata. As such, it places no restrictions on the sign of the first stage estimate across covariate strata. However, if there are few observations in each strata, then it can produce weak instruments. Furthermore, it does not readily accommodate continuous covariates. Słoczyński (2026) notes that the weighted average interpretation of the two-stage least squares estimand can be restored by reordering the instrument. Defiers for an instrument Z are compliers for the instrument $1 - Z$. Hence, an oracle with knowledge of switcher types among covariate cells could construct an instrument such that all switchers are compliers by perturbing the sign of the instrument on a cell-by-cell basis. Słoczyński (2026) shows that this reordered IV estimator identifies a weighted average of treatment effects, but leaves the investigation of its properties to future work.

In this paper, we consider a reordered IV estimator with a plug-in estimate of switcher type. The first insight of this paper is that this estimator can be oracle equivalent. As switcher type is identified by the propensity score function, we use the estimated propensity score function to

classify switchers. Given a generic estimator of the propensity score, we show that reordered IV satisfies a form of oracle equivalence. If the propensity score is estimated at the parametric rate and the instrument is relevant everywhere apart from a set of covariate cells of measure zero, then the reordered IV estimator is asymptotically equivalent to the oracle estimator where the propensity score is known. That is, estimating the switcher type has no effect on the asymptotic variance of the estimator. Intuitively, estimating switcher type affects the estimator through classification errors. Consistent estimation of the propensity score function implies that classification errors occur when the effect of the instrument on the propensity score is small. This implies that the mass of switchers in that covariate strata is small, which in turn implies that the two-stage least squares weight is small. This allows estimation errors to vanish at quicker than $\sqrt{n}$ rate.

Given an estimated propensity score function, switchers are classified by the estimated sign of the effect of the instrument on the propensity score. The second insight of this paper is that this can be leveraged to simplify estimation. We specify the propensity score as a nonparametric monotonic transformation of a linear index. As monotonic transformations do not affect rank, we need only estimate the parameters of the linear index. We show that the propensity score has this form if the random assignment assumption of Imbens and Angrist (1994) is strengthened and an additional functional form restriction is imposed. This semiparametric specification is more flexible than assuming a parametric model for the propensity score, but the parametric component can be estimated at the $\sqrt{n}$-rate under mild restrictions. We propose estimating the parameters of the linear index using the maximum rank correlation estimator of Han (1987). This approach does not directly estimate the nonparametric function, allowing us to sidestep bandwidth selection.

We show that reordered IV with a semiparametric plug-in is $\sqrt{n}$ consistent and asymptotically normal. To characterise the finite sample properties of our estimator, we conduct a simulation exercise with two processes. We specify the first process such that both compliers and defiers are present, with similar proportions of both. For this process, the sampling distribution of two-stage least squares estimates has an extremely large sample error for all sample sizes. For all but the smallest sample size, reordered IV performs well. Its sampling distribution essentially coincides with that of a quasi-oracle estimator with knowledge of the form of the propensity score function and that of an oracle with full knowledge of switcher types. The second process is specified such that there are only compliers, and two-stage least squares does not require correction. For this process, the sampling distribution of reordered IV differs only from two-stage least squares in the smallest small. These results suggest that reordered IV resolves the problems with two-stage least squares when there are many types of switcher, but does not bias estimates when switchers are of uniform type.

We implement our estimator using the data of Card (1993). We consider four versions of

reordered IV. Our preferred semiparametric approach, and three versions which specify fully parametric forms for the propensity score function. Most versions of reordered IV estimate that around 30% of observations belong to covariate cells where switchers are defiers. While two-stage least squares estimates are stable across specifications, reordered IV estimates are unstable across specifications and versions of the estimator. These results are consistent with the instrument being weak.

1.1 Related Literature

This paper builds closely on Słoczyński (2026), which introduces weak monotonicity and proposes the reordered IV estimator. More generally, this paper contributes to the literature on the estimation of average treatment effects using instrumental variables. Imbens and Angrist (1994) introduce the monotonicity assumption and show the weighted average representation of the asymptotic two-stage least squares estimand. Angrist and Imbens (1995) extend this result from binary to multivalued treatments, and introduce the saturate and control specification for the case with covariates. Vytlacil (2002) demonstrates that the identifying assumptions of Imbens and Angrist (1994) imply and are implied by a generalised Roy selection model. Building on this, Heckman and Vytlacil (2001, 2005) show how instrumental variables can identify quantities other than the average treatment effect. Kolesár (2013) compares instrumental variable estimators in the context of the heterogeneous treatment effect model with covariates. Kolesár (2013) derives conditions under which the two-stage least squares estimated corresponds to a weighted average of treatment effects, and shows that other estimators say as limited information maximum likelihood may fail to estimate a weighted average of treatment effects. More recently, Blandhol et al. (2022) show that the rich covariates assumption of Kolesár (2013) is unlikely to hold in practice and advocate using more flexible estimators in place of TSLS, such as the double debiased estimators proposed in Chernozhukov et al. (2018) and Singh and Sun (2024).

A second related strand of the literature focuses on variations of the Imbens and Angrist (1994) monotonicity assumption. This includes Klein (2010), de Chaisemartin (2017), Small et al. (2017), and Dahl et al. (2023). Klein (2010) considers the effect of random departures from monotonicity wherein the relationship between the instrument and treatment is subject to independent idiosyncratic perturbations, and characterises the resulting bias in estimates of treatment effects. de Chaisemartin (2017) shows that treatment effects are still identified in the presence of defiers as long as the distribution of compliers over treatment effects nests the distribution of defiers. Small et al. (2017) proposes the stochastic monotonicity assumption, where responses to the instrument are idiosyncratic, but units have unobservable types and monotonicity holds on average for each type. Dahl et al. (2023) assumes that compliers and defiers are both present, but only one type is

present at each point in the potential outcome distribution. This is analogous to weak monotonicity, with switcher type distinguished by potential outcomes rather than covariates. Although potential outcomes are not directly observable, under the assumptions of Dahl et al. (2023) switcher type can be identified from the joint distribution of outcomes and treatments conditional on the instrument. Dahl et al. (2023) estimate treatment effects by direct integration of nonparametric estimates of these distributions.

Outside of the instrumental variable literature, this paper is closely related to Semenova (2025). This paper considers the Lee (2009) bounds for average treatment effects when treatment affects sample participation as well as outcomes. To obtain these bounds, Lee (2009) adopts a monotonicity assumption: treatment either encourages all units to participate in the sample or discourages all units. Semenova (2025) generalises this assumption by allowing treatment to encourage participation for some covariate strata and discourage participation for others. An analogous assumption to weak monotonicity. Semenova (2025) proposes an estimator that combines debiased estimates of bounds for different covariate strata with an estimate of switcher type. The key difference between Semenova (2025) and this paper is the target. The Lee (2009) bounds related to the average treatment effect for always-takers: those who would participate in the sample irrespective of their assigned treatment. In contrast, we target the average treatment effect for switchers. The change in target leads to an estimator with different asymptotic properties.

The remainder of the paper proceeds as follows. Section 2 introduces the reordered IV estimator, and provides a simple example to illustrate. Section 3 develops the general estimator with a semiparametric plug-in. Section 4 presents a simulation exercise, Section 5 contains an application to the data of Card (1993), Section 6 briefly concludes.

Notation: I use capital letters to denote random quantities, and lower case letters to denote fixed values: $X = x$ corresponds to the event that random variable X takes value x. $|\cdot|$ and $||\cdot||$ denote the absolute value and Euclidian norms. A T superscript signifies the transpose.

2 Negative Weights and Reordered IV

This section is divided into four parts. The first summarises known results regarding compliers and defiers. The second introduces the reordered IV estimator. The third sketches its asymptotic distribution with a generic propensity score estimator. The fourth presents a simple example to illustrate the properties of reordered IV.

2.1 Switchers, Compliers, and Defiers

Let $Y \in \mathcal{Y}$ be a scalar outcome variable, $D \in \{0,1\}$ be a scalar binary treatment variable, $Z \in \{0,1\}$ be a scalar binary instrumental variable, and $X \in \mathcal{X} \subset \mathbb{R}^{d_x}$ be a length $k \geq 1$ vector of covariates including a constant. $D(z)$ denotes a potential treatment choice given $Z = z$. That is, $D(z)$ is the treatment choice that would be observed if a unit were exogenously assigned the instrument value $Z = z$. Similarly, $Y(d, z)$ denotes a potential outcome given $D = d$ and $Z = z$. We adopt the Imbens and Angrist (1994) identifying assumptions.

Assumption 1 ***Imbens and Angrist (1994) Identifying Assumptions***

(i) Instrument exclusion: $\Pr(Y(d,1) = Y(d,0)|X = x) = 1$ *for any* x.

(ii) Random assignment: $(Y(1), Y(0), D(1), D(0)) \perp\!\!\!\perp Z|X$.

(iii) Monotonicity: Given $X = x$, *either* $D(1) \geq D(0)$ *or* $D(0) \geq D(1)$ *with probability one.*

We define switchers[1] to be units whose potential treatment choice is affected by the instrumental variable in the sense that $D(1) \neq D(0)$. All switchers are either compliers or defiers. For compliers, $D(1) > D(0)$. For defiers, $D(0) > D(1)$. Units that are not switchers are either always takers ($D(0) = D(1) = 1$) or never takers ($D(0) = D(1) = 0$).

Under Assumption 1, the proportion of switchers and their type (compliers or defiers) for each X is identified. Let $p(X, Z)$ be the propensity score

$$p(x, z) = \Pr(D = 1|X = x, Z = z) = E[D|X = x, Z = z]. \tag{1}$$

Assumption 1 implies that

$$p(x,1) - p(x,0) = E[D(1)|X = x] - E[D(0)|X = x] = s(x) \Pr(D(1) \neq D(0)|X = x), \tag{2}$$

where $s(x)$ is the sign of $p(x,1) - p(x,0)$. We adopt the convention that $s(x) = 1$ whenever $p(x,1) = p(x,0)$. Hence,

$$s(x) = \begin{cases} 1 & p(x,1) - p(x,0) \geq 0, \\ -1 & p(x,1) - p(x,0) > 0. \end{cases}$$

Assumption 1 (iii) implies that if $\Pr(D(1) \geq D(0)|X = x) > 0$ then $\Pr(D(0) > D(1)|X = x) = 0$, and vice versa. Hence, $s(x)$ delineates whether the switchers for the covariate stratum $X = x$ are compliers or defiers. $\Pr(D(1) \neq D(0)|X)$ is the magnitude of $p(X,1) - p(X,0)$, and corresponds to the proportion of switchers.

[1] I am grateful to Sami Stouli for suggesting the name 'switchers'

Vytlacil (2002) shows that the Imbens and Angrist (1994) identifying assumptions imply that a selection model of the following form holds for $X = x$

$$D = \mathbb{1}\{\alpha_x + \delta_x Z \geq V_x\} \qquad\qquad V_x \perp\!\!\!\perp Z | X = x, \tag{3}$$

α_x and δ_x are constants common to all units with $X = x$, and V_x is a random variable. It follows that the propensity score can be expressed as $p(x, z) = F_{V_x}(\alpha_x + \delta_x z)$, and $s(x) = \text{sign}(\delta_x)$.

In the context of the selection model (3), $\alpha_x + \delta_x z$ is the payoff from taking treatment, and V_x as a unit specific treatment cost. Switchers at $X = x$ are compliers if the instrument increases their net payoff from treatment, and they are defiers if it decreases their net payoff. Both compliers and defiers are present if there exist covariate strata where the instrument makes treatment more attractive, and covariate strata where it makes treatment less attractive.

An alternative interpretation follows from de Chaisemartin (2017). Suppose there are many underlying types, with each type characterised by their treatment effect. de Chaisemartin (2017) shows that Assumption 1 (iii) can be weakened to the assumption that either there are more compliers than defiers of each type, or there are more defiers than compliers. In this case, there are compliers and defiers for all values of X, and $p(x, 1) - p(x, 0)$ measures the net mass of compliers. Variation in $s(x)$ implies that there some values of x with more compliers than defiers, and some with more defiers than compliers.

2.2 Reordered IV

Consider the two-stage least squares specification

$$\begin{aligned} D &= Z\delta + X^T\alpha + \varepsilon, \\ Y &= D\tau + X^T\beta + u. \end{aligned} \tag{4}$$

Given this specification, the population estimand τ_{tsls} may be written as

$$\tau_{tsls} = \frac{E[Y(Z - L[Z|X = x])]}{E[D(Z - L[Z|X = x])]}, \tag{5}$$

where $L[Z|X]$ denotes the population linear projection of Z on X.[2] Following Kolesár (2013), we impose the restriction that $L[Z|X]$ coincides with the conditional expectation of Z. Blandhol et al. (2022) refer to this assumption as 'rich covariates'.

Assumption 2 (Rich Covariates) $L[Z|X = x] = E[Z|X = x]$.

[2]Specifically, $L[Z|X] = X^T\gamma$ where $\gamma = \text{argmin}_\gamma E[(Z - X^T\gamma)^2]$.

Słoczyński (2026) shows that, under Assumptions 1 and 2, τ_{tsls} may be written as

$$\tau_{tsls} = \frac{E[s(x)\omega(x)\tau(x)]}{E[s(x)\omega(x)]}, \tag{6}$$

with

$$\begin{aligned} \omega(x) &= \text{Var}(Z|X=x)\ |p(x,1) - p(x,0)|, \\ \tau(x) &= E[Y(1) - Y(0)|D(1,x) \neq D(0,x), X = x]. \end{aligned} \tag{7}$$

For completeness, Appendix A.1 provides a derivation. $\tau(x)$ is the conditional local average treatment effect (LATE) for the $X = x$ subpopulation: an average of the treatment effects experienced by the switchers in the $X = x$ stratum. $\omega(x)$ is the proportion of switchers with $X = x$ scaled by the conditional variance of Z.

The decomposition (6) is a weighted average of $\tau(x)$ with the weights $s(x)\omega(x)/E[s(x)\omega(x)]$. If there are both compliers and defiers $s(x)$ takes values -1 and 1, and some weights are negative. This presents two issues. First, even if the sign of $\tau(x)$ is constant for all possible values of x, the sign of τ_{tsls} may differ.[3] Second, notice that the denominator of (6) can be expressed as $\Pr(s(x) = 1)E[\omega(x)|s(x) = 1] - \Pr(s(x) = -1)E[\omega(x)|s(x) = -1]$. Its magnitude is therefore determined by the difference in weights between the two types of switchers. If both types are present the denominator becomes smaller, leading to less precise estimates. In a sense, the instrument becomes weaker.

The root of the problem is that the specification (4) imposes that Z and D are monotonically related. In doing so, it gets the sign 'wrong' for one type of switcher. This problem can be resolved if we can construct an instrument such that the sign of the relationship between Z and D is constant. The reordered IV estimator proposed by Słoczyński (2026) achieves this. Consider the constructed instrument

$$Z_r = \mathbb{1}\{s(X) = 1\}Z + \mathbb{1}\{s(X) = -1\}(1 - Z)$$

Since defiers for the instrument Z are compliers for the instrument $1 - Z$, by construction, all switchers are compliers for Z_r. Under Assumptions 1 and 2, the estimand for the instrument Z_r is

$$\tau_r = \frac{E[Y s(x)(Z - E[Z|X = x])]}{E[Ds(x)(Z - E[Z|X = x])]} = \frac{E[\omega(x)\tau(x)]}{E[\omega(x)]}, \tag{8}$$

with all quantities defined as in (7). For completeness, a derivation is given in Appendix A.2. The estimand (8) is guaranteed to be a weighted average with all weights positive.

[3] In the terminology of Blandhol et al. (2022), the TSLS estimand is not weakly causal.

2.3 The Asymptotic Distribution of Reordered IV

Słoczyński (2026) considers the asymptotic reordered IV estimand in the case where $s(X)$ is known and shows that it satisfies (8). Here we consider a feasible reordered IV estimator.

To construct this estimator, we require an estimate of $s(X)$. Let $s^*(X)$ be the true sign function. Equation (2) implies that

$$s^*(X) = 2\mathbb{1}\{p^*(X,1) - p^*(X,0) \geq 0\} - 1.$$

where $p^*(X,Z)$ is the true propensity score function. Given an estimated propensity score function $\hat{p}(X,Z)$, we can define an estimated sign function $\hat{s}(X)$ as

$$\hat{s}(X) = 2\mathbb{1}\{\hat{p}(X,1) - \hat{p}(X,0) \geq 0\} - 1. \tag{9}$$

A feasible reordered IV estimator based on the sample analog of the moment condition (8) is then

$$\hat{\tau}_r = \frac{\sum_{i=1}^n \hat{s}(X_i) Y_i (Z_i - X_i^T \hat{\gamma})/n}{\sum_{i=1}^n \hat{s}(X_i) D_i (Z_i - X_i^T \hat{\gamma})/n}, \tag{10}$$

where $\hat{s}(X)$ is as defined in (9). $X^T\hat{\gamma}$ is the sample linear projection of Z on X. Under Assumption 2 this is a consistent estimator for $E[Z|X=x]$. To understand the asymptotics of the feasible estimator (10), it is helpful to define the following oracle estimator

$$\hat{\tau}_o = \frac{\sum_{i=1}^n s^*(X_i) Y_i (Z_i - X_i^T \hat{\gamma})/n}{\sum_{i=1}^n s^*(X_i) D_i (Z_i - X_i^T \hat{\gamma})/n}. \tag{11}$$

This oracle is still subject to finite sample noise and must estimate γ, but has full knowledge of $s(X)$.

The following assumptions place restrictions on the behaviour of the estimator $\hat{p}(x,z)$ and the distribution of $p^*(x,1) - p^*(x,0)$. Some additional regularity conditions are stated in Assumption 9 in Appendix B.

Assumption 3 ($\sqrt{n}$-Uniform Consistency)

$$||\hat{p} - p^*||_\infty = O_p(n^{-1/2})$$

where

$$||\hat{p} - p^*||_\infty = \sup_{x \in \mathcal{X}, z \in \{0,1\}} |\hat{p}(x,z) - p^*(x,z)|$$

Assumption 4 (VC-Class) *Define the class of functions*

$$\mathcal{S} \coloneqq \{s(X) = 2\mathbb{1}\{p(X,1) \geq p(X,0)\} - 1 : p \in \mathcal{P}\}$$

where $\mathcal{P}$ is the class of propensity score functions. $\mathcal{S}$ is a Vapnik-Chervonenkis (VC) subgraph class of functions.

Assumption 5 (Margin Condition) *There exist $C > 0$, $\eta > 0$, and $\zeta > 0$ such that for $0 < t \leq \eta$,*

$$\Pr(|p^*(x,1) - p^*(x,0)| \leq t) \leq Ct^{\zeta}.$$

Assumption 3 is a strong form of consistency. I impose this form of consistency as it enables results to be stated in a transparent manner. Assumption 4 restricts the complexity of the decision boundary induced by the propensity score functions. It rules out decision boundaries that are arbitrarily flexible.[4] This Assumption is used to deal with several empirical process type terms. Assumptions 3 and 4 hold for a variety of functional forms. For example, if covariates have bounded support, they hold for any parameteric single index model, such as probit and logit, where the index function is Lipschitz continuous. This is discussed in more detail in Section 3. Assumption 5 rules out distributions of $|p^*(x,1) - p^*(x,0)|$ that have mass points at 0, and densities that grow too rapidly in this neighbourhood. If $|p^*(x,1) - p^*(x,0)|$ has a bounded density, then Assumption 5 holds with $\zeta = 1$. Margin conditions are also used in Kitagawa and Tetenov (2018) and Semenova (2025).

The following theorem establishes the asymptotic distributions of the feasible and oracle estimators

Theorem 1 *Suppose Assumptions 1, 2 and 9 hold, then*

$$\sqrt{n}(\hat{\tau}_o - \tau_r^*) \xrightarrow{d} N(0, \sigma_{\tau_o})$$

If, in addition, Assumptions 3 - 5 hold, then

$$\sqrt{n}(\hat{\tau}_r - \hat{\tau}_o) = o_p(1)$$

where τ_r^* is the asymptotic estimand given in (8). A proof and the expression for σ_{τ_o} are given in Appendix B. This theorem states that both the feasible and oracle estimators are asymptotically normal, and that their asymptotic distributions coincide. That is, estimating $s(X)$ does not affect the asymptotic variance of the estimator.

This result can be understood as follows. The feasible and oracle estimators differ only in replacing $s^*(X)$ with its estimate $\hat{s}(X)$. In a decomposition, this leads to additional terms like $E[(\hat{s}(X) - s^*(X))D(Z - X^T\gamma^*)]$, which adjust for the plug-in estimate. These terms are bounded in magnitude by the expectation of $\mathbb{1}\{\hat{s}(X) - s^*(X)\}$ multiplied by $|p^*(X,1) - p^*(X,0)|$. For example,

$$\begin{aligned} |E[(\hat{s}(X) - s^*(X))D(Z - X^T\gamma^*)]| &= |E[(\hat{s}(x) - s^*(x))\mathrm{Var}(Z|X = x)(p^*(x,1) - p^*(x,0)]|, \\ &\leq CE[\mathbb{1}\{\hat{s}(x) \neq s^*(x)\}|p^*(x,1) - p^*(x,0)|]. \end{aligned}$$

[4] In other words, the decision boundary cannot be too 'wiggly'.

The event $\hat{s}(x) \neq s^*(x)$ is a classification error. $\hat{s}(X)$ uses estimated propensity scores to classify switchers as compliers ($\hat{s}(X) = 1$) or defiers ($\hat{s}(X) = -1$). If $\hat{s}(X)$ differs from $s^*(X)$ compliers are misclassified as defiers, or vice versa. Using the form of the sign function,

$$\hat{s}(x) \neq s^*(x) \implies |p^*(x,1) - p^*(x,0)| \leq |\hat{p}(x,1) - \hat{p}(x,0) - (p^*(x,1) - p^*(x,0))|.$$

The decision boundary for $s^*(x)$ is $p^*(x,1) - p^*(x,0) = 0$. $|p^*(x,1) - p^*(x,0)|$ measures the distance from this boundary. For missclassification to occur, the distance between $\hat{p}(x,1) - \hat{p}(x,0)$ and $p^*(x,1) - p^*(x,0)$, must exceed $|p^*(x,1) - p^*(x,0)|$. The sup norm $||\hat{p} - p^*||_\infty$ provides a bound on the distance between $\hat{p}(x,1) - \hat{p}(x,0)$ and $p^*(x,1) - p^*(x,0)$

$$|\hat{p}(x,1) - \hat{p}(x,0) - (p^*(x,1) - p^*(x,0))| \leq 2||\hat{p} - p^*||_\infty.$$

Hence,

$$\begin{aligned} E[\mathbb{1}\{\hat{s}(x) \neq s^*(x)\}|p^*(x,1) - p^*(x,0)|] &\leq E[[\mathbb{1}\{|p^*(x,1) - p^*(x,0)| \leq 2||\hat{p} - p^*||_\infty\}|p^*(x,1) - p^*(x,0)|], \\ &\leq ||\hat{p} - p^*||_\infty \Pr(|p^*(x,1) - p^*(x,0)| \leq 2||\hat{p} - p^*||_\infty), \\ &\leq C||\hat{p} - p^*||_\infty^{1+\zeta}, \end{aligned}$$

where the final line uses Assumption 5. By Assumption 3, $||\hat{p} - p^*||_\infty^{1+\zeta}$ is $O_p(n^{-(1+\zeta)/2})$, so $\sqrt{n}||\hat{p} - p^*||_\infty^{1+\zeta}$ is $o_p(1)$, and these terms vanish asymptotically.

Intuitively, the correct classification of switchers for $X = x$ does not require that the propensity score function is estimated perfectly, merely that that estimate is within a margin $|p^*(x,1) - p^*(x,0)|$. This margin is directly related to the weights in (10): the covariate values where switchers are the most difficult to classify are also those with the smallest weights. This means that, asymptotically, classification mistakes have a negligible effect on the estimand.

2.4 A Simple Example

Suppose that covariates consist of a constant and a single categorical variable X, and that Z is indepdendent of X

$$\Pr(Z = 1) = \Pr(Z = 0) = 1/2.$$

Let X take values $\{-K, ..., -1, 1, ..., K\}$. Assume that it is fixed in repeated samples: a sample of size n is comprised of $n_x = n/2K$ observations for each value of X. Consider the selection model

$$D = \mathbb{1}\{\delta_x Z \geq V\},$$

with δ_x chosen so that $p(x,1) - p(x,0) = \kappa x$, where $\kappa > 0$. For $X < 0$, switchers are defiers, for $X > 0$ switchers are compliers, and the mass of switchers increases linearly with the magnitude

of X. The true propensity score distribution has no mass at $p(x,1) = p(x,0)$. so Assumption 5 holds. Notice that $E[D(Z - E[Z|X])] = 0$, so the standard two-stage least squares estimand does not exist.

Let $U(0)$ and $U(1)$ be potential outcome residuals. The outcome process is

$$\begin{aligned} Y &= Y(0) + D(Y(1) - Y(0)), \\ Y(0) &= U(0), \\ Y(1) &= \beta + U(1). \end{aligned}$$

I assume that $(U(1), U(0), V) \perp\!\!\!\perp (X, Z)$ and $(U(1), U(0), V) \sim N(0, \Sigma)$. Σ is an arbitrary covariance matrix. For simplicity, I set the diagonal elements to 1. In the case of V, this is without loss of generality.

In this setting, the quantities defined in (7) can be explicitly calculated. For $X = x$,

$$\begin{aligned} \omega(x) &= \frac{1}{4}\kappa|x|, \\ s^*(x) &= \begin{cases} -1 & x < 0, \\ 1 & x > 0, \end{cases} \\ \tau(x) &= \beta + (\rho_{1V} - \rho_{0V}) s(x) \frac{\phi(0) - \phi(\delta_x)}{\kappa|x|}. \end{aligned}$$

where ρ_{1V} and ρ_{0V} are the correlation between $U(1)$ and V, and the correlation between $U(0)$ and V, and $\phi(\cdot)$ is the standard normal density. Using the decomposition (8), the asymptotic reordered IV estimand is

$$\frac{\sum_x \Pr(X = x)\omega(x)\tau(x)}{\sum_x \Pr(X = x)\omega(x)} = \beta + \frac{(\rho_{1V} - \rho_{0V}) \sum_x s(x)(\phi(0) - \phi(\delta_x))}{\sum_x \kappa|x|} = \beta,$$

Using this data generating process, I compare the oracle and feasible estimators. I consider cell-by-cell estimation. $E[Z|X = x]$ is estimated as the sample mean for the cell $X = x$, and $\hat{p}(x, z)$ is estimated as the sample mean of D given $X = x$ and $Z = z$.[5] It is easy to show that these estimators satisfy Assumptions 2 and 3.

I set $K = 4$, $\kappa = 0.05$. This implies that switcher shares are between 0.05 and 0.2. I use $n_x = 50$, 100, 200, and 400. The total sample size is $8n_x$. The results of Section 2.3 imply that the sampling distribution of the feasible estimator should converge to that of the oracle estimator as misclassification becomes concentrated at covariate values with few switchers. Table 1 shows misclassification probabilities for each mass of switchers given each sample size together with the probability that switchers are misclassified for at least one value of X. Even with $n_x = 400$, misclassification occurs in a little under 1/3 of draws. For simplicity, I set $B = 0$ so the asymptotic reordered IV estimand is zero. Σ is chosen randomly.

[5] If all observations for $X = x$ have the same value of Z, I set $\hat{s}(x) = 0$.

	$\kappa\|x\| = 0.05$	$\kappa\|x\| = 0.1$	$\kappa\|x\| = 0.15$	$\kappa\|x\| = 0.2$	Overall
$n_x = 50$	0.363	0.240	0.143	0.074	0.853
$n_x = 100$	0.309	0.158	0.064	0.020	0.715
$n_x = 200$	0.240	0.078	0.016	0.002	0.525
$n_x = 400$	0.158	0.022	0.001	0.000	0.324

Table 1: Probability of misclassification given n_x.

Table 2 contains summary statistics for $\hat{\tau}_r$ and $\hat{\tau}_o$ calculated from 1,000 draws for each sample size. Figure 1 plots the densities of $\sqrt{n}(\hat{\tau}_r - \tau_r)$ and $\sqrt{n}(\hat{\tau}_0 - \tau_r)$. The sampling distributions behave in line with the results of Section 2.3. For $n_x = 50$, the two are disimilar, but converge as the sample size increases. The sampling distribution of the oracle estimator is more dispersed than that of the feasible estimator, particularly in the smaller sample sizes. This arises due to negative weights. The oracle estimand is a weighted sum with nonnegative weights when the data is distributed according to the population distribution. The feasible estimator used here is, by construction, a weighted sum with nonngeative weights for any sample.[6] When $s^*(X) \neq \hat{s}(X)$, the oracle estimator includes negative weights. This decreases the magnitude of the denominator, which increases the variance of the estimator. In terms of mean squared error (MSE), the feasible estimator is preferable to the oracle estimator, but the oracle estimator exhibits less bias.

Note that this result is specific to the saturate and control specification used. In particular, it rests on $\hat{s}(X)$ being equal to the sign of the sample conditional covariance for each X, which is not an implication of Assumption 3 and does not hold for a generic propensity score estimator.

3 Semiparametric Reordered IV

The example in Section 2.4 shows that the reordered IV estimator is $\sqrt{n}$-consistent and satisfies oracle equivalence. In this example, $s(X)$ was estimated as the difference in sample conditional

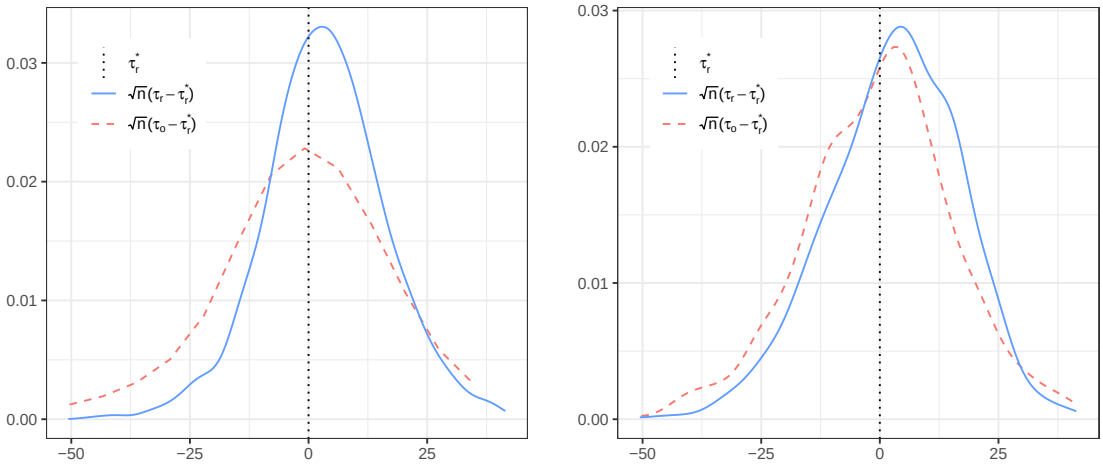
[6]The estimated sign function specified here can be written as

$$\hat{s}(x) = \text{sign}\left(\frac{\sum_i \mathbb{1}\{X_i = x\}Z_i D_i}{\sum_i \mathbb{1}\{X_i = x\}Z_i} - \frac{\sum_i \mathbb{1}\{X_i = x\}(1 - Z_i)D_i}{\sum_i \mathbb{1}\{X_i = x\}(1 - Z_i)}\right).$$

Let $\bar{Z}(x) = \sum_i \mathbb{1}\{X_i = x\}Z_i / \sum_i \mathbb{1}\{X_i = x\}$. Given that $E[Z|X]$ is estimated by $\bar{Z}(x)$, for a generic sign function $s(X)$, the denominator of the reordered IV estimator is

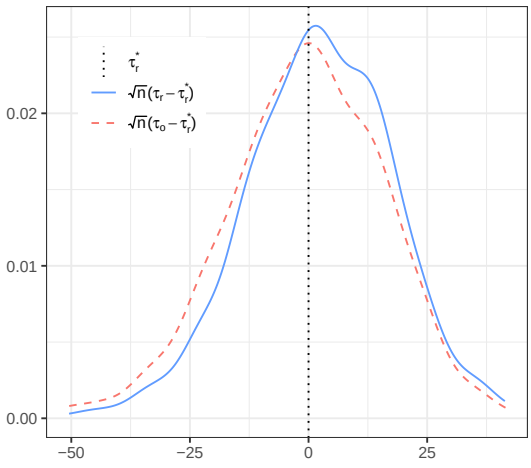
$$\frac{n_x}{n}\sum_x s(x)\bar{Z}(x)\left(1 - \bar{Z}(x)\right)\left(\frac{\sum_i \mathbb{1}\{X_i = x\}Z_i D_i}{\sum_i \mathbb{1}\{X_i = x\}Z_i} - \frac{\sum_i \mathbb{1}\{X_i = x\}(1 - Z_i)D_i}{\sum_i \mathbb{1}\{X_i = x\}(1 - Z_i)}\right).$$

If $s(x)$ coincides with $\hat{s}(x)$, then the denominator is a sum of positive terms. If $s(x) \neq \hat{s}(x)$ for some x, then the denominator includes negative terms.

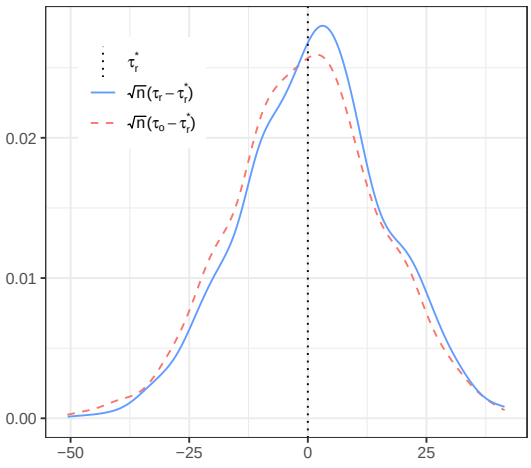


(a) $n_x = 50, n = 400$

(b) $n_x = 100, n = 800$



(c) $n_x = 200, n = 1600$



(d) $n_x = 400, n = 3200$

Figure 1: Densities of $\sqrt{n}(\hat{\tau}_r - \tau_r^*)$ and $\sqrt{n}(\hat{\tau}_o - \tau_r^*)$. n is the sample size, n_x is the number of observations for rach value of X. Densities are calculated using results from 1,000 draws and the rule-of-thumb bandwidth.

n_x	Mean	Standard Deviation	MSE	Median	First Quartile	Third Quartile
Oracle Estimator						
50	-0.069	4.578	20.967	-0.021	-0.595	0.540
100	-0.011	0.690	0.477	0.023	-0.373	0.337
200	-0.012	0.413	0.171	-0.006	-0.272	0.282
400	-0.008	0.272	0.074	0.000	-0.188	0.167
Feasible Estimator						
50	0.176	0.653	0.457	0.174	-0.233	0.568
100	0.104	0.500	0.260	0.133	-0.222	0.472
200	0.057	0.390	0.155	0.065	-0.200	0.322
400	0.023	0.266	0.071	0.028	-0.160	0.190

Table 2: Summary statistics for the sampling distribution of the oracle and feasible estimators. Results are for 1,000 runs.

means for each value of X, which is not feasible in general. In this section, we construct a reordered IV estimator where $s(X)$ is constructed using semiparametric estimates of the propensity score.

3.1 Propensity Score Functional Form

We impose a semiparametric functional form on the propensity score function.

Assumption 6 (Functional Form) *The propensity score may be written as*

$$p(X, Z) = \varphi(X^T\alpha + ZX^T\delta)$$

where $\varphi(\cdot) : \mathbb{R} \to [0, 1]$ *is an unknown strictly increasing function, and* $\alpha \in \mathcal{A} \subset \mathbb{R}^{d_x}$ *and* $\delta \in \mathcal{D} \subset \mathbb{R}^{d_x}$ *are vectors of parameters.*

This functional form specifies the propensity score as a single index model with an unknown monotonic transformation function $\varphi(\cdot)$. This functional form accommodates all parametric single index models, such as probit and logit, but is more flexible in that the $\varphi(\cdot)$ is left unknown.

An benefit of the functional form of Assumption 6 leads to a simple form for $s(X)$. Recall that $s(X) = 2\mathbb{1}\{p(x, 1) \geq p(x, 0)\} - 1$. Imposing $p(x, z) = \varphi(x^T\alpha + zx^T\delta)$,

$$p(x, 1) \geq p(x, 0) \iff x^T\delta \geq 0,$$

where we use the fact that $\varphi(\cdot)$ is a monotonic function. This implies that, to construct an estimate $\hat{s}(x)$, we require only estimates of the interaction parameters $\hat{\delta}$, the function $\varphi(\cdot)$ is irrelevant. This

is advantageous as, in semiparametric models, the parametric component is typically estimated at a faster rate than the non-parametric component.

In Section 2.3 we showed asymptotic normality and oracle equivalence of the reordered IV estimator provided that propensity score estimates satisfy uniform consistency (Assumption 3) and the decision boundary is a VC-class (Assumption 4). For this functional form, the decision boundary is $x^T\delta$, which is a finite dimensional vector space and can easily be shown to be a VC-class. As we wish to avoid estimating the nonparametric function $\varphi(\cdot)$, we modify Assumption 3. Let δ^* be the true value of δ and $\hat{\delta}$ be a generic estimator.

Assumption 7

(i) $\hat{\delta} - \delta^* = O_p(n^{-1/2})$

(ii) There exists a universal constant $C > 0$ *such that, for any* $x \in \mathcal{X}$*,*

$$|p^*(x,1) - p^*(x,0)| \leq C|X^T\delta^*|$$

(iii) For any $\epsilon > 0$ *there exists* $C > 0$ *such that* $\Pr(||X|| > C) \leq \epsilon$

Characterising the asymptotic distribution of $\hat{\tau}_r$ requires dealing with terms such as $\Pr(\hat{s}(X) \neq s(X)$. Under Assumption 3, the sup norm was used to bound these terms. Under Assumption 7 they are instead bounded by $\hat{\delta} - \delta^*$. Given Assumption 6,

$$\hat{s}(X) \neq s(X) \implies |X^T\delta^*| \leq |X^T(\delta^* - \hat{\delta})| \leq ||X|| \cdot ||\delta^* - \hat{\delta}||$$

By Assumption 7 (iii), we can find C such that $||X|| < C$ with high probability. Combining with Assumption 7, we have that if $\hat{s}(X) \neq s^*(X)$

$$|p^*(x,1) - p^*(x,0)| \leq CX^T\delta^* \leq C||\delta^* - \hat{\delta}||$$

so that

$$\Pr(\hat{s}(X) \neq s^*(X)) \leq \Pr(|p^*(x,1) - p^*(x,0)| \leq ||\delta^* - \hat{\delta}||)$$

We can then apply Assumption 5 and Assumption 7 (i) to show that this is $o_p(1)$.

3.2 Assumptions Underlying the Functional Form

Assumption 6 states the functional form as a fait accompli, but it can be related to the assumptions of Imbens and Angrist (1994). First, we need to generalise the potential outcome notation introduced in Section 2 as follows. Let $D(x,z)$ denote a potential treatment choice given $X = x$ and $Z = z$. This is the treatment status that would be observed for a unit if their instrument were

exogenously set to z *and* their covariates exogenously set to x. Analogously, let $Y(d, x, z)$ denote a potential outcome given $D = d$, $X = x$ and $Z = z$. Modify the Imbens and Angrist (1994) assumptions to

Assumption 8 (Modified Imbens and Angrist (1994) Assumptions)
Maintain Assumption 1 (i) and (iii). Replace (ii) with

(ii) Random assignment: For all $x \in \mathcal{X}$, $(Y(1,x), Y(0,x)) \perp\!\!\!\perp Z|X$ *and* $(D(x,1), D(x,0)) \perp\!\!\!\perp (X, Z)$.

Assumption (8) strengthens random assignment to joint independence of the covariates and instrument from potential treatment choices. This is weaker than full exogeneity of conditioning covariates as they may still be endogenous with respect to potential outcomes. In addition, monotonicity still does not apply to conditioning covariates, so variation in X does not identify a causal effect.

Assumption 1 does not place restrictions on how individuals assigned covariates $X = x$ would behave in the counterfactual case where their covariates are $x' \neq x$. Fix two subpopulations with $(X, Z) = (x, z)$ and $(X, Z) = (x', z')$. The counterfactual propensity scores for these subpopulations for all possible (x'', z'') are given by

$$E[D(x'', z'')|X = x, Z = z]$$
$$E[D(x'', z'')|X = x', Z = z']$$

If potential treatment choices are not independent of X and Z, these functions may not coincide. Furthermore, each function is only estimable for potential treatment choices which coincide with the assigned covariates and instrument. On the other hand, under Assumption 8, we have

$$E[D(x'', z'')|X = x, Z = z] = E[D(x'', z'')|X = x', Z = z'] = E[D(x'', z'')|X = x'', Z = z''].$$

Hence, there is a single propensity score function, and this function is identified from the data. Intuitively, Assumption 8 imposes that, in a counterfactual world where units with $X = x$ and $X = x'$ were instead assigned $X = x''$, their treatment propensity scores would be the same as the observable propensity scores from the units who are actually assigned $X = x''$.

Appendix C.2 discusses this in more detail and provides a proof. Furthermore, it shows that 8 implies that the propensity score function can be written as

$$p(x, z) = F_V(\alpha(x) + z\delta(x)),$$

where $F_V(\cdot)$ is an unknown CDF, and $\alpha(x)$ and $\delta(x)$ are unknown nonparametric functions. Assumption 6 is then obtained by restricting $\alpha(x)$ and $\delta(x)$ to be linear indexes.

3.3 Estimation of $s(X)$

As we do not require $\varphi(\cdot)$, we can use an estimator which avoids explicit estimation of this nonparametric function. We propose estimating δ using the Maximum Rank Correlation (MRC) estimator of Han (1987), a semiparametric estimator for single index models.

To introduce this estimator, let $X = (1, X_1)$ where X_1 contains all conditioning covariates apart from the constant. Write $\alpha = (\alpha_0, \alpha_1)$ and $\delta = (\delta_0, \delta_1)$ so that $X^T\alpha = \alpha_0 + X_1^T\alpha_1$ and $ZX^T\delta = \delta_0 Z + Z(X^T\delta_1)$. Define the normalised parameter vector

$$(\tilde{\alpha}_1, \tilde{\delta}_0, \tilde{\delta}_1) \coloneqq \frac{(\alpha_1, \delta_0, \delta_1)}{||(\alpha_1, \delta_0, \delta_1)||}$$

Let n_1 be the number of observations with $D = 1$, and n_0 the number with $D = 0$. The MRC estimator is

$$(\hat{\tilde{\alpha}}_1, \hat{\tilde{\delta}}_0, \hat{\tilde{\delta}}_1) = \text{argmax}_{\tilde{\alpha}_1, \tilde{\delta}_0, \tilde{\delta}_1} \binom{N}{2}^{-1} \sum_{i=1}^{n_1} \sum_{j=1}^{n_0} \mathbb{1}\{X_{1i}^T\tilde{\alpha}_1 + Z_i\tilde{\delta}_0 + Z_i X_{1i}^T\tilde{\delta}_1 > X_{1j}^T\tilde{\alpha}_1 + Z_j\tilde{\delta}_0 + Z_j X_{1j}^T\tilde{\delta}_1\}.$$

This criterion sums over all pairs of observations i, j such $D_i = 1$ and $D_j = 0$. A pair contributes 1 to the criterion if $X_{1i}^T\tilde{\alpha}_1 + Z_i\tilde{\delta}_0 + Z_i X_{1i}^T\tilde{\delta}_1 > X_{1j}^T\tilde{\alpha}_1 + Z_j\tilde{\delta}_0 + Z_j X_{1j}^T\tilde{\delta}_1$, and 0 otherwise. The total contribution of any observation i with $D_i = 1$ is equal to its rank among all n_0 observations with $D_j = 0$. $(\hat{\tilde{\alpha}}_1, \hat{\tilde{\delta}}_0, \hat{\tilde{\delta}}_1)$ is chosen so that no change in value would increase the average rank of $D_i = 1$ observations among the $D_j = 0$ observations. This estimator does not identify α_0 as adding a constant to the index does not change ranks. Similarly, multiplying all parameters by a positive constant does not change ranks, so parameters are only identified up to scale. To construct $\hat{s}(x)$, we require only the sign of $x^T\delta$, which is unchanged if we multiply δ by a positive constant. Hence, estimating $\tilde{\delta} = (\tilde{\delta}_0, \tilde{\delta}_1)$ is sufficient.

Han (1987) proves that the MRC estimates of $(\tilde{\alpha}_1, \tilde{\delta}_0, \tilde{\delta}_1)$ are consistent, and Sherman (1993) that they are $\sqrt{n}$-consistent and asymptotically normal. Subbotin (2007) shows that standard errors can be obtained using the standard nonparametric bootstrap. The MRC estimator does not explicitly compute the nonparametric function $\varphi(\cdot)$, which simplifies estimation. On the other hand, the criterion function (3.3) is neither smooth nor globally concave. This means that the maximisation must be performed numerically by an algorithm that does not require a gradient, and that the criterion may have multiple local maxima. Shin and Todorov (2021) show that the maximisation problem can be formulated as a mixed integer programming problem, and the global maximum obtained using a branch and bound algorithm. We use a simplex method with multiple starting points.

3.4 Estimator Algorithm and Asymptotic Distribution

The general feasible reordered IV estimator can be calculated as follows.

1. Obtain estimates of the parameters $\hat{\tilde{\delta}}$ using the maximum rank correlation estimator. Calculate the estimated sign function as

$$\hat{s}(X_i) = 2\mathbb{1}\{X_i^T \hat{\tilde{\delta}} \geq 0\} - 1$$

2. Regress Z on X and save the fitted values $X^T\hat{\gamma}$.

3. Obtain $\hat{\tau}_r$ using the analogue estimator

$$\hat{\tau}_r = \frac{\sum_{i=1}^n Y_i \hat{s}(X_i)(Z_i - X_i^T\hat{\gamma})}{\sum_{i=1}^n D_i \hat{s}(X_i)(Z_i - X_i^T\hat{\gamma})}.$$

Standard errors can be obtained by nonparametric bootstrap.

The following Theorem characterises the asymptotic distribution of $\hat{\tau}_r$ with a generic estimator of $\hat{\delta}$. The result is identical to Theorem 1, but Assumptions 3 and 4 are replaced by Assumptions 6 and 7.

Theorem 2 *Suppose Assumptions 1, 2, 5 - 7 and 9 hold. Then*

$$\sqrt{n}(\hat{\tau}_r - \hat{\tau}_o) = o_p(1)$$

A proof is presented in Appendix B. Showing that this theory applies when we use the maximum rank correlation estimator requires verifying the assumptions. The only assumption which restricts the estimator is Assumption 7. A minor issue is that Assumption 7 (ii) is stated in terms of δ, the maximum rank correlation estimator instead estimates scaled parameters $\tilde{\delta}$. However, the constant in 7 (ii) can be redefined without and the assumption restated for scaled parameters. Assumption 7 (i) is shown in Sherman (1993). Hence, the Theorem 2 applies when we use the maximum rank correlation estimator.

4 Simulations

Theorem 2 characterises the asymptotic behaviour of the general feasible reordered IV estimator. In this section, we evaluate its finite sample performance through a simulation exercise.

Maintaining the notation of Section 3.3, let $X = (1, X_1)$ and say that X_1 consists of 3 variables.

The data generating process is

$$
\begin{aligned}
&X \sim N(0, I) \qquad Z \sim \text{binomial}(1/2), \\
&D = \mathbb{1}\{\alpha_0 + X_1^T \alpha_1 + Z(\delta_0 + X_1^T \delta_1) \geq V\}, \\
&Y(0) = X_1^T \beta_0 + U(0), \\
&Y(1) = \upsilon + X_1^T \beta_1 + U(1), \\
&Y = DY(1) + (1 - D)Y(0), \\
&(V, U(1), U(0)) \sim N(0, \Sigma).
\end{aligned} \tag{12}
$$

Here, the proportions of always takers, never takers, and switchers vary with X. Consider the subpopulation of units with $X = x$. The switchers in this subpopulation are compliers if $\delta_0 + x^T \delta_1 > 0$. In this case, a fraction $\Phi(\alpha_0 + x^T \alpha_1)$ of the subpopulation are always takers, $\Phi((\alpha_0 + \delta_0) - x^T(\alpha_1 + \delta_1)) - \Phi(\alpha_0 + x^T \alpha_1)$ are compliers, and the remainder are never takers. If $\delta_0 + x^T \delta_1 < 0$, then the switchers are defiers. In this case, $\Phi((\alpha_0 + \delta_0) - x^T(\alpha_1 + \delta_1))$ corresponds to the share of always takers, $\Phi(\alpha_0 + x^T \alpha_1) - \Phi((\alpha_0 + \delta_0) - x^T(\alpha_1 + \delta_1))$ to the share of defiers, and the remainder are never takers. If $\delta_0 \neq 0$ and $\delta_1 \neq 0$, then the hyperplane $\delta_0 + X^T \delta_1 = 0$ defines a set of X values where the instrument is irrelevant. As X is unbounded, there will exist both compliers and defiers as long as $\delta_1 \neq 0$, although one group can be arbitrarily small if δ_1 is small in magnitude relative to δ_0.

We consider two parameterisations. One in which all switchers are compliers, and one in which there is a mix of compliers and defiers. Instead of directly setting α_1, δ_1 and β_1, we parameterise the joint distribution of $(X^T \alpha_1, X^T \delta_1, X^T \beta_1)$. This imposes a set of constraints on $(\alpha_1, \delta_1, \beta_1)$. In each simulation draw, we generate $(\alpha_1, \delta_1, \beta_1)$ under these constraints.

Parameter values that are common across both parameterisations are presented in Table 3. The average treatment effect for the process (12) is $\zeta_0 - \beta_0$. Holding this difference fixed, any values of β_0 and ζ_0 will generate the same population distribution of treatment effects. For simplicity, we set $\beta_0 = 0$, and $\zeta_0 = 0.1$. The standard deviation of $X^T \beta_1$ is set to 0.45, with $X^T \beta_1$ uncorrelated with $X^T \alpha_1$ and $X^T \delta_1$. All diagonal elements of Σ are set to 1, the correlation between V and $U(1)$ is set to -0.6, and the correlation between V and $U(0)$ is set to 0.4.

Parameter values which differ between the two parameterisations are presented in Table 4. α_0, δ_0, the standard deviation of $X^T \alpha_1$, the standard deviation of $X^T \delta_1$, and the covariance of $X^T \delta_1$ and $X^T \alpha$ are jointly chosen, and their values change across parameterisations. When there are both compliers and defiers, we fix the total (i.e integrating over X and Z) shares of always takers, compliers and defiers to be 0.4, 0.11, and 0.09. We then specify that the standard deviation of $p(X, 0)$ is 0.15 and the standard deviation of $p(X, 1)$ is 0.1. When there are only compliers, $\delta = 0$

Parameter	Value
β_0	0
ζ_0	0.100
$\beta^T\beta$	0.450
$\beta^T\delta$	0
$\beta^T\alpha$	0

Parameter	Value
σ_0	1
σ_1	1
ρ_{1v}	-0.6
ρ_{0v}	0.4
ρ_{01}	0

Table 3: Common Monte Carlo Parameter Values

Parameter	Specification 1	Specification 2
α_0	-0.027	-0.27
δ_0	0.053	0.549
$\alpha^T\alpha$	0.405	0.419
$\delta^T\delta$	0.652	0
$\alpha^T\delta$	-0.261	0

Table 4: Monte Carlo Parameter Values

and we need only three conditions. We fix the total share of always takers to be 0.4, the share of compliers to be 0.2, and the standard deviation of $p(X, 0)$ to be 0.15.

We consider sample sizes $n = 100, 500, 1000$ and 2000. For each sample size, we draw 1000 sets of parameters, and generate a sample for each. We estimate the treatment effect by two-stage least squares, reordered IV, and two oracle versions of reordered IV. The two-stage least squares estimate is calculated as

$$\hat{\tau}_{tsls} = \frac{\sum_{i=1}^n Y_i(Z_i - X_i^T\hat{\gamma}_n)}{\sum_{i=1}^n D_i(Z_i - X_i^T\hat{\gamma}_n)}$$

where $X^T\hat{\gamma}_n$ is the finite sample linear projection of Z on X. As Z is independent of X, this is a consistent estimate of $E[Z|X]$.

For reordered IV, we specify the propensity score as $\varphi(\alpha_0 + X^T\alpha + Z(\delta_0 + X^T\delta))$ with $\varphi(\cdot)$ an unknown function, and estimate $\tilde{\alpha}_1$, $\tilde{\delta}_0$ and $\tilde{\delta}_1$ by the maximum rank correlation estimator. The reordered IV estimate is then calculated as

$$\hat{\tau}_r = \frac{\sum_{i=1}^n Y_i \operatorname{sign}(X_i^T\hat{\tilde{\delta}}_1 + \hat{\tilde{\delta}}_0)(Z_i - X_i^T\hat{\gamma}_n)}{\sum_{i=1}^n D_i \operatorname{sign}(X_i^T\hat{\tilde{\delta}}_1 + \hat{\tilde{\delta}}_0)(Z_i - X_i^T\hat{\gamma}_n)}$$

To calculate this estimator, we use the Nelder-Mead algorithm with multiple starting points including the probit, logit, and linear probability model estimates of α_1, δ_0 and δ_1[7], random perturbations

[7]The estimates are normalised to be length 1 before being passed to the solver.

of these estimates, and a set of random points.

We also calculate quasi-oracle and oracle versions of the reordered IV estimates. Estimates for the quasi-oracle are calculated as above, but δ_0 and δ_1 are estimated by probit. That is, this oracle is knows the correct functional form for $\varphi(\cdot)$, but must still estimate parameters. For the oracle version, the sign function is evaluated using the true parameter values δ_0 and δ_1

$$\hat{\tau}_{oracle} = \frac{\sum_{i=1}^{n} Y_i \operatorname{sign}(X_i^T\delta_1 + \delta_0)(Z_i - X_i^T\hat{\gamma}_n)}{\sum_{i=1}^{n} D_i \operatorname{sign}(X_i^T\delta_1 + \delta_0)(Z_i - X_i^T\hat{\gamma}_n)}$$

Note that, when there are only compliers, the $\hat{\tau}_{oracle}$ coincides with $\hat{\tau}_{tsls}$.

Table 5 presents results for the process with both compliers and defiers. For each estimator and sample size, it shows the mean across runs, the standard error, and selected quantiles of the distribution. Estimates obtained by two-stage least squares are extremely noisy for all sample sizes. For $n = 100$, the reordered IV estimates are noisy, but the standard error shrinks rapidly as the sample size grows. All versions of reordered IV perform similarly for all but the smallest sample size. Figure 2 plots the densities of the reordered IV and oracle reordered IV densities for $n = 500, n = 1,000$ and $n = 2,000$. The densities almost perfectly coincide.

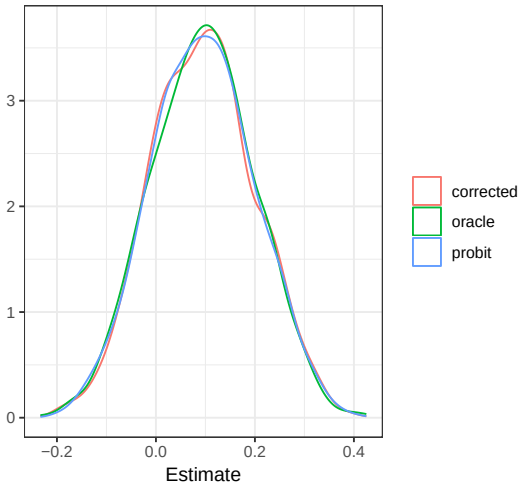
Table 6 and Figure 3 repeat this for the process with compliers only. The pattern is similar. Reordered IV is noisy in the smallest sample, but performs similarly to both normal two-stage least squares and the quasi-oracle in larger samples.

5 Empirical Application

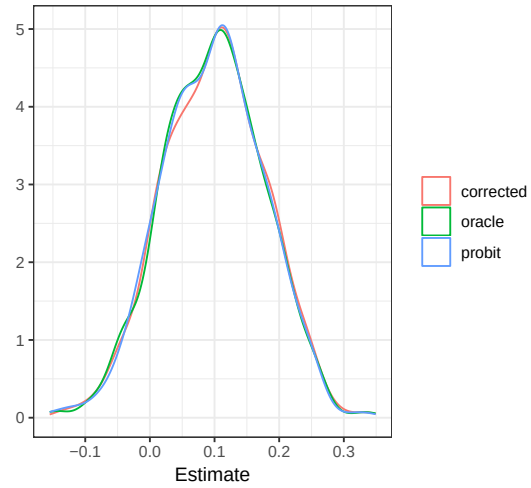
In this section, we apply the estimator to the design of Card (1993), which estimates the returns to education. The outcome Y is log hourly wage in 1976. D is years of education. I binarise D with the threshold set to 16 years of education. Z is a binary variable that equals one for individuals who lived close to an accredited four-year college in 1966.[8] Baseline controls are potential experience (defined as age$-$years of education-6), a race dummy, and dummies for residence in the south and a standard metropolitan statistical area (SMSA) in 1976. A second specifications adds a dummy for residence in the South in 1966, and fixed effects for region of residence in 1966. The third further controls for each parents years of education. The fourth introduces a categorical variable constructed from both parents years of education. The final specification then adds controls for family structure at age 14.

The sample consists of 3,010 observations. When all conditioning covariates are included, there are 2,101 unique covariate combinations in the sample. As such, estimating quantities for individual covariate cells is infeasible.

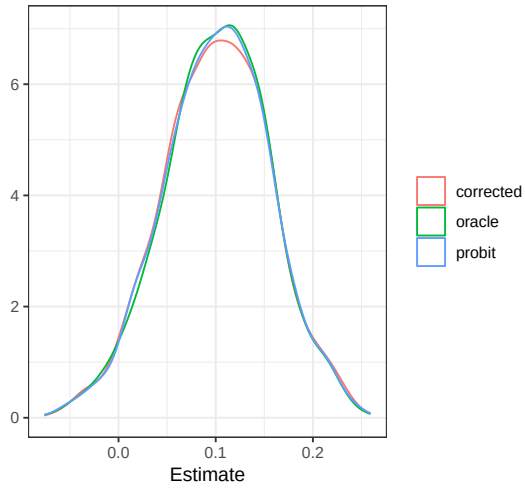
[8] Specifically, if their local labour market in 1966 contained an accredited four year college.



(a) $n = 500$



(b) $n = 1000$

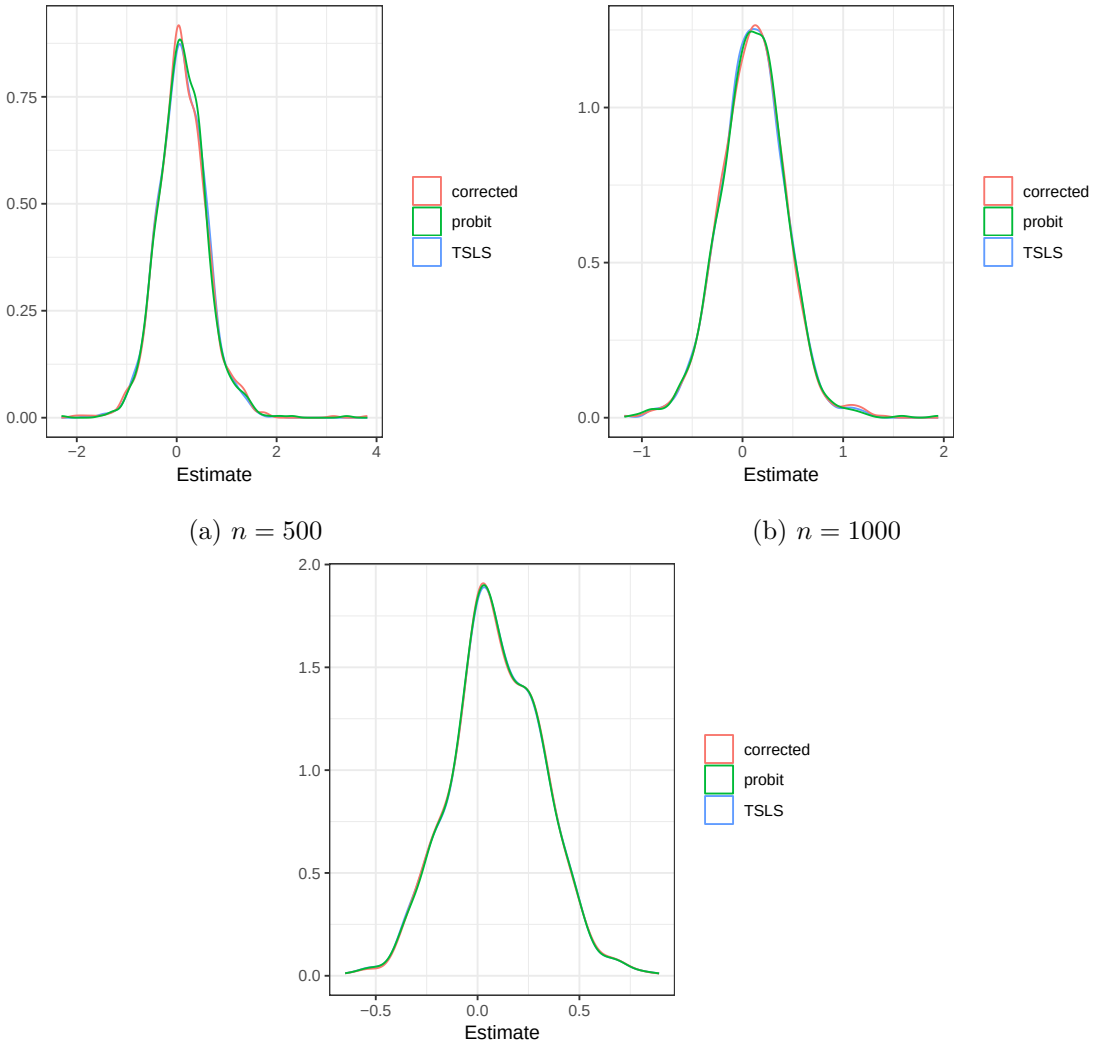


(c) $n = 2000$

Figure 2: Simulation Results with $\delta \neq 0$. n is the sample size. The panels show the densities of τ_r and τ_o calculated with the rule-of-thumb bandwidth and results from 1,000 samples.

n	Mean	Standard Error	0.1	0.25	0.5	0.75	0.9
TSLS							
100	-0.317	61.516	-6.063	-1.761	0.115	2.100	6.070
500	0.788	87.109	-5.286	-1.683	0.063	1.812	4.904
1000	1.574	79.575	-4.417	-1.699	0.084	1.836	5.447
2000	-0.424	26.976	-3.632	-1.257	0.147	1.664	4.664
Reordered IV (Semiparametric)							
100	1.348	43.707	-1.246	-0.459	0.129	0.678	1.292
500	0.122	0.501	-0.465	-0.195	0.123	0.410	0.702
1000	0.104	0.339	-0.311	-0.113	0.117	0.325	0.512
2000	0.107	0.243	-0.185	-0.061	0.112	0.261	0.423
Reordered IV (Probit)							
100	-0.102	3.577	-1.282	-0.513	0.129	0.702	1.403
500	0.118	0.527	-0.499	-0.202	0.102	0.435	0.758
1000	0.103	0.351	-0.321	-0.122	0.108	0.338	0.533
2000	0.105	0.241	-0.186	-0.063	0.107	0.273	0.408
Reordered IV (Oracle)							
100	-0.384	20.685	-1.812	-0.650	0.086	0.825	1.859
500	0.138	0.606	-0.523	-0.202	0.116	0.453	0.822
1000	0.109	0.352	-0.332	-0.115	0.100	0.348	0.535
2000	0.107	0.247	-0.198	-0.061	0.106	0.267	0.419

Table 5: Summary Statistics with $\delta \neq 0$. The mean, standard deviation and selected quantiles of estimates over 1,000 runs. 0.1 is the first decile, 0.25 the first quartile, 0.5 the median, and so on.



(a) $n = 500$

(b) $n = 1000$

(c) $n = 2000$

Figure 3: Simulation Results with $\delta = 0$. n is the sample size. The panels show the densities of τ_c and τ_{oracle} calculated with the rule-of-thumb bandwidth and results from 1,000 samples.

n	Mean	Standard Error	0.1	0.25	0.5	0.75	0.9
TSLS							
100	-0.399	14.510	-1.584	-0.655	0.087	0.744	1.590
500	0.112	0.481	-0.478	-0.201	0.098	0.424	0.676
1000	0.099	0.322	-0.308	-0.099	0.098	0.301	0.496
2000	0.096	0.225	-0.201	-0.046	0.080	0.253	0.384
Reordered IV (Semiparametric)							
100	0.153	5.402	-1.161	-0.495	0.121	0.738	1.635
500	0.116	0.514	-0.474	-0.207	0.080	0.412	0.708
1000	0.102	0.328	-0.307	-0.104	0.104	0.304	0.474
2000	0.097	0.224	-0.196	-0.044	0.081	0.254	0.380
Reordered IV (Probit)							
100	1.231	41.681	-1.211	-0.519	0.077	0.641	1.613
500	0.117	0.496	-0.477	-0.199	0.104	0.417	0.689
1000	0.102	0.329	-0.307	-0.103	0.099	0.303	0.508
2000	0.096	0.224	-0.199	-0.046	0.080	0.253	0.384

Table 6: Summary Statistics with $\delta = 0$. The mean, standard deviation and selected quantiles of estimates over 1,000 runs. 0.1 is the first decile, 0.25 the first quartile, 0.5 the median, and so on.

I estimate each specification by two-stage least squares, and by four variations of reordered IV. The first three variations use parametric specifications for the propensity score: probit, logit and linear probability model. The final variation uses the semiparametric specification proposed in Section 3. In each case, the estimated parameters of the linear index are used to construct $\hat{s}(X)$. The semiparametric estimates are calculated as follows. The Nelder-Mead simplex algorithm is used with a total of 38 starting points. The estimates from the probit, logit, and linear probability models, five random perturbations of each, and twenty random points. Standard errors for the reodered IV estimates are obtained by bootstrap.

Results are presented in Table 7. Each column corresponds to estimates for one set of controls. Standard errors are in paretheses. The two-stage least squares estimates are stable across specifications. For the second and fifth specifications, estimates are significant at the 10% level. In contrast, the reordered IV estimates are unstable across specifications, and none are significant. Regardless of the variation used, estimates for the first two specifications are imprecise.

Estimator					
TSLS	1.786	1.392*	1.460	1.574	1.544*
	(1.147)	(0.836)	(0.893)	(0.959)	(0.889)
Probit	3.645	1.106	-1.870	2.457	-0.426
	(20.480)	(7.127)	(2.970)	(3.073)	(2.970)
Logit	-0.391	-0.494	-1.151	0.792	-0.689
	(66.010)	(5.260)	(2.956)	(3.185)	(3.053)
LPM	-11.906	-2.122	-3.666	0.232	-2.856
	(25.612)	(15.198)	(4.239)	(3.413)	(3.604)
Semiparametric	2.425	-0.046	-3.439	0.839	-1.591
	(21.221)	(129.660)	(3.048)	(2.993)	(3.150)
Region Controls	No	Yes	Yes	Yes	Yes
Parental Education	No	No	Yes	Yes	Yes
Interacted Parental Education	No	No	No	Yes	Yes
Family Composition	No	No	No	No	Yes

Table 7: Estimates

Probit	0.163	0.271	0.307	0.293	0.300
Logit	0.186	0.259	0.290	0.277	0.286
LPM	0.245	0.322	0.338	0.373	0.370
Reordered IV	0.078	0.278	0.269	0.278	0.302

Table 8: Shares

6 Conclusion

This paper studies the reordered IV estimator suggested in Słoczyński (2026), which allows for both compliers and defiers to be present. Implementing this estimator requires estimating for each covariate value whether switchers are compliers or defiers. We showed that, if the instrument is sufficiently strong in the sense that a margin condition holds, then this estimation is costless in terms of asymptotic variance. We proposed a general estimator where switcher type is estimated semiparametrically, and showed that it is $\sqrt{n}$-consistent and asymptotically normal. In a simulation exercise, we found that the reordered IV estimator performs well, and resolves the negative weights problem encountered by conventional two-stage least squares. We applied the estimator to the data of Card (1993), and found evidence of both defiers and compliers. The difference between the conventional two-stage least squares and reordered IV estimates was large.

The methods presented in this paper could be further developed in two ways. First, they can be extended to multivalued and continuous treatments and instruments. Second, this paper has adopted the rich covariates assumption that the expectation of Z is a linear function of X. Modern estimators, such as that proposed in Chernozhukov et al. (2018) relax this assumption and use flexible machine learning methods to approximate the expectation function. Incorporating these flexible methods into the reordered IV estimator may be a promising direction for future research.

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A Omitted Derivations

A.1 Formula for τ_{tsls}

The denominator of (5) is decomposed as follows

$$\begin{aligned}
E[D(Z - L[Z|X=x])] &= E[D(Z - E[Z|X=x])] \\
&= E[\Pr(Z=1|X=x)(1-E[Z|X=x])E[D|X=x,Z=1] \\
&\quad - \Pr(Z=0|X=x)E[Z|X=x]E[D|X=x,Z=0]] \\
&= E[\mathrm{Var}(Z|X=x)(E[D(1)|X=x,Z=1] - E[D(0)|Z=0,X=x])] \\
&= E[\mathrm{Var}(Z|X=x)(E[D(1)|X=x] - E[D(0)|X=x])] \\
&= E[\mathrm{Var}(Z|X=x)(E[D(1)-D(0)|X=x])] \\
&= E[s(x)\omega(x)]
\end{aligned}$$

The first equality uses Assumption 2. The second uses the law of iterated expectations. The third uses that Z is binary, so $E[Z|X] = \Pr(Z=1|X)$ and $\Pr(Z=1|X)\Pr(Z=0|X) = \mathrm{Var}(Z|X)$. The fourth equality follows from Assumption 1 (ii). As $D(Z)$ and Z are independent conditional on

X, $E[D(z)|Z=z, X=x] = E[D(z)|X=x]$. The final equality holds by the definitions of $\omega(x)$ and $s(x)$. Under Assumption 1 (iii) $E[D(1)-D(0)|X=x]$ is either $\Pr(D(1)>D(0)|X=x)$ or $-\Pr(D(0)>D(1)|X=x)$, which is equivalent to $\Pr(D(1)\neq D(0)|X=x)$ multiplied by $s(x)$.

For the numerator of (5)

$$\begin{aligned}
E[Y(Z-L[Z|X=x])] &= E[Y(Z-E[Z|X=x])] \\
&= E[(Y(0)+D(Y(1)-Y(0)))(Z-E[Z|X=x])] \\
&= E[E[D(Y(1)-Y(0))(Z-E[Z|X=x]|X=x]] \\
&= E[\mathrm{Var}(Z|X=x)(E[D(Y(1)-Y(0))|X=x,Z=1] \\
&\qquad - E[D(Y(1)-Y(0))|X=x,Z=0])] \\
&= E[\mathrm{Var}(Z|X=x)E[(D(1)-D(0))(Y(1)-Y(0))|X=x]] \\
&= E[s(x)\omega(x)\tau(x)]
\end{aligned}$$

The first equality uses Assumption 2. The second follows as $Y = DY(1)+(1-D)Y(0)$. The third holds as $E[Y(0)(Z-E[Z|X=x])|X=x] = E[Y(0)|X=x]E[Z-E[Z|X=x]|X=x] = 0$, where we use the conditional independence of $Y(0)$ and Z. For the fifth equality, we have $E[D(Y(1)-Y(0))|X=x,Z=z] = E[D(z)(Y(1)-Y(0))|X=x,Z=z]$, which conditional independence then implies is equal to $E[D(z)(Y(1)-Y(0))|X=x]$. By Assumption 1 (iii), $E[(D(1)-D(0))(Y(1)-Y(0))|X=x]$ is either $\Pr(D(1)>D(0)|X=x)E[Y(1)-Y(0)|X=x, D(1)>D(0)]$ or $-\Pr(D(0)>D(1)|X=x)E[Y(1)-Y(0)|X=x, D(0)>D(1)]$, which is equivalent to $\Pr(D(1)\neq D(0)|X=x)\tau(x)$ multiplied by $s(x)$.

A.2 Derivation of τ_r

Consider the denominator of (8)

$$\begin{aligned}
E[Ds(x)(Z-L[Z|X=x])] &= E[Ds(x)(Z-E[Z|X=x])] \\
&= E[\Pr(Z=1|X=x)s(x)(1-E[Z|X=x])E[D|X=x,Z=1] \\
&\qquad - \Pr(Z=0|X=x)s(x)E[Z|X=x]E[D|X=x,Z=0]] \\
&= E[\mathrm{Var}(Z|X=x)s(x)(E[D(1)|X=x,Z=1]-E[D(0)|Z=0,X=x])] \\
&= E[\mathrm{Var}(Z|X=x)s(x)(E[D(1)|X=x]-E[D(0)|X=x])] \\
&= E[\mathrm{Var}(Z|X=x)s(x)(E[D(1)-D(0)|X=x])] \\
&= E[s(x)^2\omega(x)] \\
&= E[\omega(x)]
\end{aligned}$$

The penultimate equality uses that $E[D(1)-D(0)|X=x] = \Pr(D(1)>D(0)|X=x) - \Pr(D(1)< D(0)|X=x) = s(x)|p(x,1)-p(x,0)|$.

For the numerator of (8),

$$\begin{aligned}
E[Ys(x)(Z-L[Z|X=x])] &= E[Ys(x)(Z-E[Z|X=x])] \\
&= E[(Y(0)+D(Y(1)-Y(0)))s(x)(Z-E[Z|X=x])] \\
&= E[E[D(Y(1)-Y(0))(Z-E[Z|X=x]|X=x]] \\
&= E[\mathrm{Var}(Z|X=x)s(x)(E[D(Y(1)-Y(0))|X=x,Z=1] \\
&\qquad - E[D(Y(1)-Y(0))|X=x,Z=0])] \\
&= E[\mathrm{Var}(Z|X=x)s(x)E[(D(1)-D(0))(Y(1)-Y(0))|X=x]] \\
&= E[s(x)^2\omega(x)\tau(x)] \\
&= E[\omega(x)\tau(x)]
\end{aligned}$$

B Proofs

B.1 Additional Assumptions

The following Assumption collects regularity conditions needed for the proof of Theorem 1.

Assumption 9 *(i) The data (X,Y,D,Z) is an iid sample.*

(ii) $E[Ds^*(X)(Z-X^T\gamma^*)] > 0$

(iii) $E[Y|X=x,Z=z]$ *and* $E[Y^2|X=X,Z=z]$ *are finite for all* x *and* z.

(iv) $\Sigma_X = E[XX^T]$ *is positive definite and* $E[||X||^2]$ *is finite.*

(v) $\gamma \in \Gamma$ *where* Γ *is a compact subset of* $\mathbb{R}^{d_x}$.

(vi) The moment $E[|Y|\cdot||X||]$ *is finite.*

(vii) For any $t\in[0,1]$, *the expectation of* $||X||$ *conditional on* $|\Delta p^*(X)| \le t$ *is bounded.*

As $E[Y|X=x,Z=1]-E[Y|X=x,Z=0] = \omega(x)\tau(x)$, Assumption 9 (iii) implies that $\tau(x)$ is bounded.

Similarly, Assumption 9 (vi) implies that τ_r^* is bounded in magnitude.

$$|\tau_r^*| = \left|\frac{E[Ys^*(x)(Z-X^T\gamma^*)]}{E[Ds^*(x)(Z-X^T\gamma^*)]}\right| \le \frac{E[|Ys^*(x)(Z-X^T\gamma^*)|]}{|E[Ds^*(x)(Z-X^T\gamma^*)]|} \le \frac{E[|Y|]+CE[|Y|\cdot||X||]]}{|E[Ds^*(x)(Z-X^T\gamma^*)]|}.$$

Hence, It follows that $\tau_r \in \mathcal{T}_r$ where $\mathcal{T}_r \subset \mathbb{R}$ is a compact set.

B.2 Notation

Let $E_n[\cdot]$ be the empirical expectation operator and $\mathbb{G}_n(\cdot)$ be the empirical process operator. For a generic variable X, $E_n[X] = \sum_{i=1}^n X_i/n$ and $\mathbb{G}_n(X) = \sqrt{n}(E_n(X) - E(X))$.$C$ is used to denote a genetic constant, whose value is not fixed.

Define the errors

$$e_Y = Y - \tau_r^* D,$$
$$e_Z = Z - X^T\gamma^*,$$

and the moment conditions,

$$m^{e_Y}(s,\gamma) = e_Y s(X)(Z - X^T\gamma),$$
$$m^D(s,\gamma) = Ds(X)(Z - X^T\gamma).$$

Notice that

$$\tau_r^* = \frac{E[Ys^*(X)(Z - X^T\gamma^*)]}{E[Ds^*(X)(Z - X^T\gamma^*)]} \implies 0 = \frac{E[e_Y s^*(X)(Z - X^T\gamma^*)]}{E[Ds^*(X)(Z - X^T\gamma^*)]} \implies E[m^{e_Y}(s^*,\gamma^*)] = 0. \quad (13)$$

Finally, write

$$\Delta p(x) = p(x,1) - p(x,0).$$

B.3 Proof of Theorem 1

Under Assumption 2, $\hat{\gamma}$ is the linear projection of Z on X. Hence, $(\hat{\gamma} - \gamma^*) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{e}_Z$, where $\mathbf{X}$ is a matrix formed by stacking X and $\mathbf{e}_Z$ is a vector that collects e_Z. From Assumption 9 (i) and (iv),

$$\sqrt{n}(\hat{\gamma} - \gamma^*) = \Sigma_X^{-1}\mathbf{X}^T\mathbf{e}_Z/\sqrt{n} + o_p(1), \quad (14)$$

where $\mathbf{X}^T\mathbf{e}_Z/\sqrt{n} = O_p(1)$ by the Lindeberg-Feller CLT.

Now consider the oracle estimator. $\sqrt{n}(\hat{\tau}_o - \tau_r^*)$ can be decomposed as

$$\begin{aligned}\sqrt{n}(\hat{\tau}_o - \tau_r^*) &= \sqrt{n}\frac{E_n[m^{e_Y}(s^*,\hat{\gamma})]}{E_n[m^D(s^*,\hat{\gamma})]}, \\ &= \frac{\mathbb{G}_n[m^{e_Y}(s^*,\hat{\gamma})] + \sqrt{n}E[m^{e_Y}(s^*,\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)]}{E_n[m^D(s^*,\hat{\gamma})]},\end{aligned}$$

where the second line uses (13). Lemma 1 establishes the convergence of the empirical process $\mathbb{G}_n(m^{e_Y}(s^*,\hat{\gamma})$ to $\mathbb{G}_n(m^{e_Y}(s^*,\hat{\gamma}^*)$. Using this, the form of $m^{e_Y}(s,\gamma)$, and (14),

$$\sqrt{n}(\hat{\tau}_r - \tau_r^*) = \frac{\mathbb{G}_n(m^{e_Y}(s^*,\gamma^*)) - E[e^Y s^*(X)X]^T\Sigma_X^{-1}\mathbf{X}^T\mathbf{e}_Z/\sqrt{n}}{E_n[m^D(s^*,\hat{\gamma})]} + o_p(1) \quad (15)$$

Furthermore, using (13) and the definition of e_Z

$$\mathbb{G}_n(m^{e_Y}(s^*,\gamma^*)) = \sqrt{n}\left(E_n[m^{e_Y}(s^*,\gamma^*)] - E[m^{e_Y}(s^*,\gamma^*)]\right) = \sqrt{n}E_n[e_Y s^*(X)e_Z],$$

so

$$\sqrt{n}(\hat{\tau}_r - \tau_r^*) = \frac{\sqrt{n}E_n[e_Y s^*(X)e_Z] - E[e_Y s^*(X)X]^T\Sigma_X^{-1}\mathbf{X}^T\mathbf{e}_Z/\sqrt{n}}{E_n[m^D(s^*,\hat{\gamma})]} + o_p(1). \tag{16}$$

The vector $(e_Y s^*(X)e_Z, X^T e_Z)$ is mean zero and has covariance matrix

$$\Omega = E\left[\begin{pmatrix} e_Y s^*(X)e_Z \\ X^T e_Z \end{pmatrix}\begin{pmatrix} e_Y s^*(X)e_Z \\ X^T e_Z \end{pmatrix}^T\right]$$

By the Cramer-Wold device and the Lindeberg-Feller CLT, $(e_Y s^*(X)e_Z, X^T e_Z)$ is normally distributed. By Lemma 2, $E_n[m^D(s^*,\hat{\gamma})] \xrightarrow{p} E[m^D(s^*,\gamma^*)]$. Applying Slutsky's Theorem to (16),

$$\sqrt{n}(\hat{\tau}_r - \tau_r^*) \xrightarrow{d} N(0,\sigma_{\tau_r})$$

with

$$\sigma_{\tau_r}^2 = E[m^D(s^*,\gamma^*)]^{-2}\begin{pmatrix} 1 \\ -E[e_Y s^*(X)X]^T\Sigma_X^{-1} \end{pmatrix}^T \Omega \begin{pmatrix} 1 \\ -E[e_Y s^*(X)X]^T\Sigma_X^{-1} \end{pmatrix}$$

For the feasible estimator, $\sqrt{n}(\hat{\tau}_r - \tau_r^*)$, can be decomposed as

$$\sqrt{n}(\hat{\tau}_r - \tau_r^*) = \frac{\mathbb{G}_n(m^{e_Y}(\hat{s},\hat{\gamma})) + \sqrt{n}E[m^{e_Y}(\hat{s},\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)]}{E_n[m^D(\hat{s},\hat{\gamma})]}$$

By Lemma 3, $\sqrt{n}(E[m^D(\hat{s},\hat{\gamma}) - m^D(s^*,\gamma^*)]) = \sqrt{n}E[Ds^*(X)X]^T(\hat{\gamma}-\gamma^*) + o_p(1)$ and $\sqrt{n}(E[m^{e_Y}(\hat{s},\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)]) = \sqrt{n}E[e_Y s^*(X)X]^T(\hat{\gamma}-\gamma^*) + o_p(1)$. Lemma 5 shows that $\mathbb{G}_n(m^{e_Y}(\hat{s},\hat{\gamma})) = \mathbb{G}_n(m^{e_Y}(s^*,\gamma^*)) + o_p(1)$. Hence

$$\sqrt{n}(\hat{\tau}_r - \tau_r^*) = \frac{\left(\mathbb{G}_n(m^{e_Y}(s^*,\gamma^*)) - E[e_Y s^*(X)X]^T\Sigma_X^{-1}\mathbf{X}^T\mathbf{e}_Z/\sqrt{n}\right)}{E_n[m^D(\hat{s},\hat{\gamma})]}$$

which differs from the oracle estimator only in that $E[m^D(\hat{s},\hat{\gamma})]$ replaces $E[m^D(s^*,\hat{\gamma})]$. The result follows by the same argument as the oracle estimator and Lemma 6.

Lemma 1 *Suppose that Assumptions 1, 2, and 9 hold*

(i) $\mathbb{G}_n(m^{e_Y}(s^*,\hat{\gamma}) = \mathbb{G}_n(m^{e_Y}(s^*,\gamma^*)) + o_p(1)$

(ii) $\mathbb{G}_n(m^D(s^*,\hat{\gamma}) = \mathbb{G}_n(m^D(s^*,\gamma^*)) + o_p(1)$

For claim (i),

$$\begin{aligned}
&\mathbb{G}_n(m^{e_Y}(s^*,\hat{\gamma})) - \mathbb{G}_n(m^{e_Y}(s^*,\gamma^*)) \\
&= \sqrt{n}\left(E_n[m^{e_Y}(s^*,\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)] - E[m^{e_Y}(s^*,\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)]\right) \\
&= \sqrt{n}(E_n[e_Y s^*(X)X] - E[e_Y s^*(X)X])^T(\gamma^* - \hat{\gamma})
\end{aligned}$$

$E_n[e_Y s^*(X)X] - E[e_Y s^*(X)X]$ is an average of independent mean zero vectors. Assumption (9) (ii) implies that $E[Y|X]$ and $E[Y^2|X]$ are finite. Hence,

$$\begin{aligned}
&E[e_Y s^*(X)X] = E[(E[Y|X] - \tau_r^* E[D|X])s^*(X)X] < \infty, \\
&E[e_Y s^*(X)X(e_Y s^*(X)X)^T] = E[(E[Y|X] - \tau_r^* E[D|X])^2|X]XX^T] \leq CE[XX^T] < \infty,
\end{aligned}$$

where we also use Assumption 9 (iv). This establishes that $E_n[e_Y s^*(X)X] - E[e_Y s^*(X)X]$ has a finite covariance matrix. By the Lindeberg-Levy CLT, $\sqrt{n}(E_n[Ds^*(X)X] - E[Ds^*(X)X])$ is $O_p(1)$. The claim follows as $||\hat{\gamma} - \gamma^*|| = o_p(1)$.

The argument for claim (ii) follows the same steps. In this case, we apply a CLT to $\sqrt{n}(E_n[Ds^*(X)X] - E[Ds^*(X)X])$. Since $|Ds^*(X)X| \leq |X|$, finiteness of the mean and covariance matrix are immediate in this case.

Lemma 2 *Suppose that Assumptions 1, 2, and 9 hold*

$$E_n[m^D(s^*,\hat{\gamma})] = E[m^D(s^*,\gamma^*)] + o_p(1)$$

$$\begin{aligned}
E_n[m^D(s^*,\hat{\gamma})] - E[m^D(s^*,\gamma^*)] &= E_n[m^D(s^*,\hat{\gamma}) - E[m^D(s^*,\hat{\gamma})]] + E[m^D(s^*,\hat{\gamma}) - m^D(s^*,\gamma^*)] \\
&= n^{-1/2}\mathbb{G}_n(m^D(s^*,\hat{\gamma})) + n^{-1/2}\sqrt{n}E[m^D(s^*,\hat{\gamma}) - m^D(s^*,\gamma^*)] \\
&= n^{-1/2}\mathbb{G}_n(m^D(s^*,\gamma^*)) + n^{-1/2}E[Ds^*(X)X]^T\sqrt{n}(\gamma^* - \hat{\gamma}) + o_p(1) \\
&= n^{-1/2}O_p(1) + n^{-1/2}O_p(1) + o_p(1) \\
&= o_p(1)
\end{aligned}$$

The third line uses the result of Lemma 1. For the fourth, $\mathbb{G}_n(m^D(s^*,\gamma^*))$ is $O_p(1)$ by the Lindeberg-Levy CLT, and $\sqrt{n}(\hat{\gamma} - \gamma^*)$ is $O_p(1)$ by the Lindeberg-Feller CLT.

Lemma 3 *Suppose that Assumptions 1 - 5 and 9 hold*

(i) $\sqrt{n}E[m^{e_Y}(\hat{s},\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)] = -\sqrt{n}E[e_Y s^*(X)X]^T(\hat{\gamma} - \gamma^*) + o_p(1)$

(ii) $\sqrt{n}E[m^D(\hat{s},\hat{\gamma}) - m^D(s^*,\gamma^*)]) = -\sqrt{n}E[Ds^*(X)X]^T(\hat{\gamma} - \gamma^*) + o_p(1)$

For claim (i),

$$\sqrt{n}E[m^{e_Y}(\hat{s},\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)] = \sqrt{n}E[m^{e_Y}(s^*,\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)] - \sqrt{n}\mathcal{A}_1 + \sqrt{n}\mathcal{A}_2$$

with

$$\begin{aligned}\mathcal{A}_1 &= E[e_Y(\hat{s}(X) - s^*(X))X^T(\hat{\gamma} - \gamma^*)],\\ \mathcal{A}_2 &= E[e_Y(\hat{s}(X) - s^*(X))(Z - X^T\gamma^*)],\end{aligned}$$

Each of these terms is $o_p(1)$. For $\mathcal{A}_1$,

$$\begin{aligned}|\mathcal{A}_1| &= |E[E[e_Y|X](\hat{s}(X) - s^*(X))X^T(\hat{\gamma} - \gamma^*)]|\\ &\leq E[|E[e_Y|X](\hat{s}(X) - s^*(X))X^T(\hat{\gamma} - \gamma^*)|]\\ &\leq CE[\mathbb{1}\{\hat{s}(X) \neq s^*(X)\}|X^T(\hat{\gamma} - \gamma^*)|]\\ &\leq CE[\mathbb{1}\{|\Delta p^*(X)| \leq 2||\hat{p} - p^*||_\infty\}||X||] \cdot ||\hat{\gamma} - \gamma^*||\\ &\leq C\Pr(|\Delta p^*(x)| \leq 2||\hat{p} - p^*||_\infty)||\hat{\gamma} - \gamma^*||\\ &\leq C||\hat{p} - p^*||_\infty^\zeta||\hat{\gamma} - \gamma^*||\\ &= O_p(n^{-(1+\zeta)/2}).\end{aligned}$$

The third line holds as $E[e_Y|X]$ is bounded. For the fourth, $X^T(\hat{\gamma}-\gamma^*)$ is a scalar, so $|X^T(\hat{\gamma}-\gamma^*)| = ||X^T(\hat{\gamma} - \gamma^*)||$. The Cauchy-Schwarz inequality then gives the inequality. The fifth line uses Assumption 9 (vii). As $\zeta > 0$, $\sqrt{n}|\mathcal{A}_1| = O_p(n^{-\zeta/2}) = o_p(1)$.

For $\mathcal{A}_2$,

$$\begin{aligned}|\mathcal{A}_2| &= |E[e_Y(\hat{s}(X) - s^*(X))(Z - X^T\gamma^*)]|\\ &= |E[(\hat{s}(x) - s^*(x))\mathrm{Var}(Z|X = x)\Delta p^*(x)(\tau(x) - \tau_r^*)]|\\ &\leq CE[\mathbb{1}\{\hat{s}(x) \neq s^*(x)\}|\Delta p^*(x)|]\\ &\leq CE[\mathbb{1}\{|\Delta p^*(x)| \leq 2||\hat{p} - p^*||_\infty\}|\Delta p^*(x)|]\\ &\leq C||\hat{p} - p^*||_\infty \Pr(|\Delta p^*(x)| \leq 2||\hat{p} - p^*||_\infty)\\ &\leq C||\hat{p} - p^*||_\infty^{1+\zeta}\\ &= O_p(n^{-(1+\zeta)/2}).\end{aligned}$$

The fourth line holds as $\tau(x)$ is bounded by Assumption 9. This implies that $\sqrt{n}\mathcal{A}_2 = O_p(n^{-\zeta/2}) = o_p(1)$.

For claim (ii), the decomposition is similar. The additional terms are

$$\begin{aligned}\mathcal{B}_1 &= E[D(\hat{s}(X) - s^*(X))X^T(\hat{\gamma} - \gamma^*)]\\ \mathcal{B}_2 &= E[D(\hat{s}(X) - s^*(X))(Z - X^T\gamma^*)]\end{aligned}$$

For $\mathcal{B}_1$,

$$|\mathcal{B}_1| = |E[D(\hat{s}(X) - s^*(X))X^T(\hat{\gamma} - \gamma^*)]| \leq E[|\mathbb{1}\{\hat{s}(X) \neq s^*(X)\}|X^T(\hat{\gamma} - \gamma^*)|].$$

The remaining steps are identical to the proof for $\mathcal{A}_1$.

For $\mathcal{B}_2$,

$$|\mathcal{B}_2| = |E[D(\hat{s}(X) - s^*(X))(Z - X^T\gamma^*)]| \leq E[\mathbb{1}\{\hat{s}(X) \neq s^*(X)\}|\Delta p^*|]$$

The claim follows by the same argument as $\mathcal{A}_2$.

Lemma 4 *Suppose that Assumptions 2, 4, and 9 hold.*

(i) Consider the pseudometric

$$\rho^{e_Y}((s,\gamma),(s',\gamma')) = E[(m^{e_Y}(s,\gamma) - m^{e_Y}(s',\gamma'))^2]^{1/2}$$

For every $\eta > 0, \epsilon > 0$, there exists $\xi > 0$ such that

$$\lim_{n\to\infty} \Pr\left(\sup_{(s,s')\in\mathcal{S}^2,(\gamma,\gamma')\in\Gamma^2:\rho^{e_Y}((s,\gamma),(s',\gamma'))<\xi} |\mathbb{G}_n(m^{e_Y}(s,\gamma) - \mathbb{G}_n(m^{e_Y}(s',\gamma'))| > \eta \right) < \epsilon$$

(ii) Consider the pseudometric

$$\rho^{D}((s,\gamma),(s',\gamma')) = E[(m^{D}(s,\gamma) - m^{D}(s',\gamma'))^2]^{1/2}$$

For every $\eta > 0, \epsilon > 0$, there exists $\xi > 0$ such that

$$\lim_{n\to\infty} \Pr\left(\sup_{(s,s')\in\mathcal{S}^2,(\gamma,\gamma')\in\Gamma^2:\rho^{D}((s,\gamma),(s',\gamma'))<\xi} |\mathbb{G}_n(m^{D}(s,\gamma) - \mathbb{G}_n(m^{D}(s',\gamma'))| > \eta \right) < \epsilon$$

Both claims can be shown by verifying Assumptions A, B, and C of Theorem 1 of Andrews (1994). Assumption C is satisfied as the data are i.i.d. To verify Assumptions A and B for claim (i), consider the function class

$$\mathcal{M}^{e_Y} \coloneqq \{m^{e_Y}(s,\gamma) : s \in \mathcal{S}, \gamma \in \Gamma\},$$

Notice that

$$|(Y - \tau_r^* D)s(X)(Z - X^T\gamma)| = |Y - \tau_r^* D||Z - X^T\gamma| \leq (|Y| + C)(1 + ||X||C)$$

where we use that Γ is compact and τ_r^* is finite. Hence, this function class possesses a measurable envelope. Assumption B then holds as the data are i.i.d. For Assumption A, any $m^{e_Y}(s,\gamma)$ can be

written as $f \times s$ for some f in the function class $\mathcal{F} \coloneqq \{(Y - \tau_r^* D)(Z - X^T\gamma) : \gamma \in \Gamma\}$ and $s \in \mathcal{S}$, with $\mathcal{S}$ as defined in Assumption 4. $\mathcal{F}$ is a finite-dimensional vector space of measurable functions so, by Lemma 2.6.16 of Vaart and Wellner (2023), it is a VC-subgraph class of functions. $\mathcal{S}$ is a VC-class by Assumption 4. By Theorem 2.6.7 of Vaart and Wellner (2023), the uniform covering numbers of $\mathcal{S}$ and $\mathcal{F}$ are finite, so they satisfy Assumption A. Theorem 3 of Andrews (1994) then implies that $m^D(s, \gamma)$ satisfies Assumption A.

For claim (ii), consider the function class

$$\mathcal{M}^D = \{m^D(s, \gamma) : s = \mathbb{1}\{p(x, 1) > p(x, 0)\}, p \in \mathcal{P}, \gamma \in \Gamma\}.$$

The envelope for this function class is $(1 + ||X||C)$, and Assumptions A and B can be verified in the same manner.

Lemma 5 *Suppose that Assumptions 1 - 5 and Assumption 9 hold.*

(i) $\mathbb{G}_n(m^{e_Y}(\hat{s}, \hat{\gamma})) = \mathbb{G}_n(m^{e_Y}(s^*, \gamma^*)) + o_p(1)$

(ii) $\mathbb{G}_n(m^D(\hat{s}, \hat{\gamma})) = \mathbb{G}_n(m^D(s^*, \gamma^*)) + o_p(1)$

For claim (i), consider $E[(m^{e_Y}(\hat{s}, \hat{\gamma}) - m^{e_Y}(s^*, \gamma^*))^2]$.

$$\begin{aligned}
E[(m^{e_Y}(\hat{s}, \hat{\gamma}) - m^{e_Y}(s^*, \gamma^*))^2] &= E[e_Y^2(\hat{s}(X)X^T(\gamma^* - \hat{\gamma}) + (\hat{s}(X) - s^*(X))(Z - X^T\gamma^*))^2] \\
&\leq CE[(\hat{s}(X)X^T(\gamma^* - \hat{\gamma}))^2] + CE[((\hat{s}(X) - s^*(X))(Z - X^T\gamma^*))^2] \\
&\leq CE\left[||X||^2\right] ||\gamma - \hat{\gamma}||^2 + CE\left[\mathbb{1}\{\hat{s}(X) \neq s(X)\}\left(Z - X^T\gamma^*\right)^2\right] \\
&\leq C||\gamma - \hat{\gamma}||^2 + CE\left[\mathbb{1}\{|\Delta p^*(x)| \leq 2||\hat{p} - p||_\infty\}\left(Z - X^T\gamma^*\right)^2\right] \\
&\leq C||\gamma - \hat{\gamma}||^2 + CE\left[\mathbb{1}\{|\Delta p^*(x)| \leq 2||\hat{p} - p^*||_\infty\}\right] \\
&\leq C||\gamma - \hat{\gamma}||^2 + C||\hat{p} - p^*||_\infty^\zeta \\
&= o_p(1).
\end{aligned}$$

The first inequality holds as Assumption 9 implies that $E[e_Y^2|X]$ is bounded. For the fifth inequality, D and Z are binary, and $X^T\gamma^* = E[Z|X] \in [0, 1]$, so $|D(Z - X^T\gamma^*)| \leq 1$. The last line holds as $||\hat{\gamma} - \gamma^*|| = o_p(1)$, $||\hat{p} - p||_\infty = o_p(1)$ and $\zeta > 0$. This implies that $\rho^D((\hat{s}, \hat{\gamma}), (s^*, \gamma^*)) = o_p(1)$, where $\rho^{e_Y}(\cdot, \cdot)$ is a pseudometric introduced in Lemma 4. The remaining steps are a standard argument.

Fix $\eta > 0$ and $\epsilon > 0$,

$$\begin{aligned}
&\Pr(|\mathbb{G}_n(m^{e_Y}(\hat{s},\hat{\gamma})) - \mathbb{G}_n(m^{e_Y}(s^*,\gamma^*))| > \eta) \\
&\leq \Pr(\rho^{e_Y}((\hat{s},\hat{\gamma}),(s^*,\gamma^*)) \geq \xi) + \Pr(|\mathbb{G}_n(m^{e_Y}(\hat{s},\hat{\gamma})) - \mathbb{G}_n(m^{e_Y}(s^*,\gamma^*)|) > \eta, \rho^{e_Y}((\hat{s},\hat{\gamma}),(s^*,\gamma^*)) < \xi) \\
&\leq \Pr(\rho^{e_Y}((\hat{s},\hat{\gamma}),(s^*,\gamma^*)) \geq \xi) \\
&\quad + \Pr\left(\sup_{(s,s')\in\mathcal{S}^2,(\gamma,\gamma')\in\Gamma^2:\rho^{e_Y}((s,\gamma),(s',\gamma'))<\xi} |\mathbb{G}_n(m^{e_Y}(s,\gamma) - \mathbb{G}_n(m^{e_Y}(s',\gamma'))| > \eta\right).
\end{aligned}$$

The first inequality holds by the union bound. By the above argument and Lemma 4, there exist ξ and N such that, for all $n > N$, both terms are smaller than $\epsilon/2$. As this argument holds for any η and ϵ, $\mathbb{G}_n(m^{e_Y}(\hat{s},\hat{\gamma})) - \mathbb{G}_n(m^{e_Y}(s^*,\gamma^*))$ is $o_p(1)$.

For claim (ii), as D is binary,

$$\begin{aligned}
E[(m^D(\hat{s},\hat{\gamma}) - m^D(s^*,\gamma^*))^2] &= E[D^2(\hat{s}(X)X^T(\gamma^* - \hat{\gamma}) + (\hat{s}(X) - s^*(X))(Z - X^T\gamma^*))^2] \\
&\leq E[(\hat{s}(X)X^T(\gamma^* - \hat{\gamma}) + (\hat{s}(X) - s^*(X))(Z - X^T\gamma^*))^2],
\end{aligned}$$

and the rest of the argument proceeds as above.

Lemma 6 *Suppose that Lemmas 3 and 5, and Assumption 9 holds*

$$E_n[m^D(\hat{s},\hat{\gamma})] = E[m^D(s^*,\gamma^*)] + o_p(1)$$

$$\begin{aligned}
E_n[m^D(\hat{s},\hat{\gamma})] - E[m^D(s^*,\gamma^*)] &= E_n[m^D(\hat{s},\hat{\gamma}) - E[m^D(\hat{s},\hat{\gamma})]] + E[m^D(\hat{s},\hat{\gamma}) - m^D(s^*,\gamma^*)] \\
&= n^{-1/2}\mathbb{G}_n(m^D(\hat{s},\hat{\gamma})) + n^{-1/2}\sqrt{n}E[m^D(\hat{s},\hat{\gamma}) - m^D(s^*,\gamma^*)] \\
&= n^{-1/2}\mathbb{G}_n(m^D(s^*,\gamma^*)) - n^{-1/2}\sqrt{n}E[Ds^*(X)X]^T(\hat{\gamma} - \gamma^*) + o_p(1) \\
&= n^{-1/2}O_p(1) + n^{-1/2}O_p(1) + o_p(1) \\
&= o_p(1)
\end{aligned}$$

The third line uses Lemmas 3 and 5. For the fourth, $\mathbb{G}_n(m^D(s^*,\gamma^*))$ is $O_p(1)$ by the Lindeberg-Levy CLT, and $\sqrt{n}(\hat{\gamma} - \gamma^*)$ is $O_p(1)$ by the Lindeberg-Feller CLT.

B.4 Proof of Theorem 2

To prove Theorem 2, we first show that Assumption 6 implies a class of sign functions which satisfy Assumption 4. Under Assumption 6,

$$\mathcal{S} \coloneqq \{s(X) = 1 - 2\mathbb{1}\{-X^T\delta > 0\} : \delta \in \mathcal{D}\}.$$

To show this is a VC-class, we apply Lemmas 2.6.16 and 2.6.20 of Vaart and Wellner (2023). The class of functions $\mathcal{F} := \{-X^T\delta : \delta \in \mathcal{D}\}$ is a finite dimensional vector space, which is a VC-subgraph class by Lemma 2.6.16. By Lemma 2.6.20 (iii), (v) and (vi), passing $\mathcal{F}$ through an indicator function with threshold zero, adding a fixed function, and multiplying by a fixed function preserve this property. We thus conclude that $\mathcal{S}$ is a VC-subgraph class.

Lemmas 7 and 8 show that the results of Lemmas 3 and 5 hold with Assumptions 6 and 7 replacing Assumption 3. The results of Lemmas 1, 4 and 6 do not directly rely on Assumption 3, nor does the main body of the proof of Theorem 1. Hence, Theorem 2 can be shown by the same argument with the new lemmas replacing the old.

Lemma 7 *Suppose that Assumptions 1, 2, 5 - 7 and 9 hold.*

(i) $\sqrt{n}E[m^{e_Y}(\hat{s},\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)] = -\sqrt{n}E[e_Y s^*(X)X]^T(\hat{\gamma}-\gamma^*) + o_p(1)$

(ii) $\sqrt{n}E[m^{D}(\hat{s},\hat{\gamma}) - m^{D}(s^*,\gamma^*)]) = -\sqrt{n}E[Ds^*(X)X]^T(\hat{\gamma}-\gamma^*) + o_p(1)$

Let ξ_n be a sequence of positive constants that grow at rate $\zeta/2$. For claim (i), proceed as in Lemma 3, obtaining the same decomposition and defining $\mathcal{A}_1$ and $\mathcal{A}_2$ identically. We verify that $\sqrt{n}\mathcal{A}_1$ and $\sqrt{n}\mathcal{A}_2$ are still $o_p(1)$. For $\mathcal{A}_1$,

$$
\begin{aligned}
|\mathcal{A}_1| &\leq CE[\mathbb{1}\{\hat{s}(X) \neq s^*(X)\}|X^T(\hat{\gamma}-\gamma^*)|] \\
&\leq CE[\mathbb{1}\{|x^T\delta^*| \leq |x^T(\hat{\delta}-\delta^*)|\}||x||] \cdot ||\hat{\gamma}-\gamma^*|| \\
&\leq CE[\mathbb{1}\{|x^T\delta^*| \leq ||x|| \cdot ||\hat{\delta}-\delta^*||\}||x||] \cdot ||\hat{\gamma}-\gamma^*|| \\
&\leq C\left(\xi_n \cdot E[\mathbb{1}\{|x^T\delta^*| \leq C\xi_n \cdot ||\hat{\delta}-\delta^*||\}] \cdot ||\hat{\gamma}-\gamma^*|| + \Pr(||x|| \geq \xi_n)||\hat{\gamma}-\gamma^*||\right) \\
&\leq C\left(\xi_n \cdot \Pr(|\Delta p^*| \leq C \cdot \xi_n \cdot ||\hat{\delta}-\delta^*||\}) \cdot ||\hat{\gamma}-\gamma^*|| + \Pr(||x|| \geq \xi_n)||\hat{\gamma}-\gamma^*||\right) \\
&\leq \xi_n^{1+\zeta} C||\hat{\delta}-\delta^*||\}^{\zeta} \cdot ||\hat{\gamma}-\gamma^*|| + C\Pr(||x|| \geq \xi_n)||\hat{\gamma}-\gamma^*|| \\
&= O_p(n^{-(1+\zeta^2)/2}) + o_p(n^{-1/2})
\end{aligned}
$$

where the last equality holds as $\Pr(||x|| \geq \xi_n) = o(1)$. As $\zeta > 0$, $\sqrt{n}\mathcal{A}_1 = o_p(1)$.

For $\mathcal{A}_2$,

$$\begin{aligned}
|\mathcal{A}_2| &\leq CE[\mathbb{1}\{\hat{s}(x) \neq s^*(x)\}|\Delta p^*(x)|] \\
&\leq CE[\mathbb{1}\{|x^T\delta^*| \leq |x^T(\hat{\delta} - \delta^*)|\}|\Delta p^*(x)|] \\
&\leq CE[\mathbb{1}\{|\Delta p^*| \leq ||x|| \cdot ||\hat{\delta} - \delta^*||\}|\Delta p^*(x)|] \\
&\leq CE[\mathbb{1}\{|\Delta p^*| \leq ||x|| \cdot ||\hat{\delta} - \delta^*||\} \cdot ||x||] \cdot ||\hat{\delta} - \delta^*|| \\
&\leq C\left(E[\mathbb{1}\{|\Delta p^*| \leq \xi_n||\hat{\delta} - \delta^*||\}] \cdot \xi_n||\hat{\delta} - \delta^*|| + \Pr(||x|| \geq \xi_n)||\hat{\delta} - \delta^*||\right) \\
&\leq C(\xi_n||\hat{\delta} - \delta^*||)^{(1+\zeta)/2} + C\Pr(||x|| \geq \xi_n)||\hat{\delta} - \delta^*|| \\
&= O_p(n^{-(1+\zeta^2)/2} + o_p(n^{-1/2})
\end{aligned}$$

which implies that $\sqrt{n}\mathcal{A}_2 = o_p(1)$. The proof of claim (ii) follows its proof in Lemma 3 with the same additional steps as above.

Lemma 8 *Suppose that Assumptions 1, 2, 5 - 7 and 9 hold.*

(i) $\mathbb{G}_n(m^{e_Y}(\hat{s}, \hat{\gamma})) = \mathbb{G}_n(m^{e_Y}(s^*, \gamma^*)) + o_p(1)$

(ii) $\mathbb{G}_n(m^{D}(\hat{s}, \hat{\gamma})) = \mathbb{G}_n(m^{D}(s^*, \gamma^*)) + o_p(1)$

For claim (i), consider $E[(m^{e_Y}(\hat{s}, \hat{\gamma}) - m^{e_Y}(s^*, \gamma^*))^2]$. Repeating the steps of Lemma 5, the only term that must be dealt with differently is $E\left[\mathbb{1}\{\hat{s}(X) \neq s(X)\}\left(Z - X^T\gamma^*\right)^2\right]$. We show that this remains $o_p(1)$ under the new assumptions. Let ξ_n be an increasing sequence of positive constants

$$\begin{aligned}
E\left[\mathbb{1}\{\hat{s}(X) \neq s(X)\}\left(D(Z - X^T\gamma^*)\right)^2\right] &\leq E\left[\mathbb{1}\{|x^T\delta^*| \leq |x^T(\hat{\delta} - \delta^*)|\}\left(Z - X^T\gamma^*\right)^2\right] \\
&\leq E\left[\mathbb{1}\{|x^T\delta^*| \leq |x^T(\hat{\delta} - \delta^*)|\}\right] \\
&= \Pr(|x^T\delta^*| \leq |x^T(\hat{\delta} - \delta^*)|, |\Delta p^*| \leq C|x^T\delta^*|) \\
&\leq \Pr(|\Delta p^*| \leq C||x|| \cdot ||\hat{\delta} - \delta^*||) \\
&\leq \Pr(|\Delta p^*| \leq C\xi_n \cdot ||\hat{\delta} - \delta^*||) + \Pr(||x|| > \xi_n) \\
&\leq C||\hat{\delta} - \delta^*||^{\zeta} + \Pr(C\xi_n||\hat{\delta} - \delta^*|| > \eta) + \Pr(||x|| > \xi_n)
\end{aligned}$$

The third line uses Assumption 7 (ii). In the final line, η is as defined in Assumption 5. Under Assumption 7, $||\hat{\delta} - \delta^*|| = O_p(n^{-1/2})$. This implies that, with $\zeta > 0$, the first term is $o_p(1)$. Furthermore, we can construct ξ_n such that $\xi_n \to \infty$ and $\xi_n||\hat{\delta} - \delta^*|| = o_p(1)$. Such a choice renders the last two terms $o(1)$. It follows that $E\left[\mathbb{1}\{\hat{s}(X) \neq s(X)\}\left(D(Z - X^T\gamma^*)\right)^2\right] = o_p(1)$. The proof can then proceed as in Lemma 5. The argument for claim (ii) is identical.

C Implied Selection Model

Consider the case where X is a single binary variable. Let $D_i(x, z)$ denote the potential treatment choice for unit i, which corresponds to their treatment choice if they are exogenously assigned covariate value x and instrument value z. Unit i's observed choice is $D_i(X_i, Z_i)$.

Theorem 3 *Let $X \in \{0, 1\}$ and $Z \in \{0, 1\}$ be a binary conditioning covariate and instrument. The Imbens and Angrist (1994) indentifying assumptions imply that potential outcomes are generated by a selection model of the form*

$$D_i(X, Z) = \mathbb{1}\{\alpha_0 + \delta_0 Z + a_i X + bXZ \geq V_{0i}\} \qquad (a_i, V_{0i}) \perp\!\!\!\perp Z|X$$

Intuitively, the Imbens and Angrist (1994) monotonicity assumption implies that, once we fix X, the effect of Z on the propensity score is uniform across units. Hence, δ_0 and b do not vary with i. Monotonicity places no restrictions on how the effect varies with X, so the actual values of δ_0 and b can be completely different.

On the other hand, the coefficient on X, a_i, is unit specific. Monotonicity does not extend to the covariates, so the effect of X need not be uniform across units. Fixing $Z = 0$ and changing $X = 0$ to $X = 1$, there can exist units with positive a_i who switch into treatment and units with negative a_i who switch out of treatment.

Turning to the unobservables, the exogeneity assumptions implies that they are independent of Z. However, conditioning covariates may not be exogenous, so may be related to these unobservables.

C.1 Proof

Vytlacil (2002) shows that, for any covariate value x, the Imbens and Angrist (1994) identifying assumptions imply and are implied by a selection model of the form

$$D_i(x, Z) = \mathbb{1}\{\alpha_x + \delta_x Z \geq V_{ix}\} \qquad V_{ix} \perp\!\!\!\perp Z|X \tag{17}$$

where V_{ix} is a selection unobservable.

Recall that the Imbens and Angrist (1994) identifying assumptions do not extend monotonicity to the covariates. This means that a change in covariate values can induce simultaneous flows into and out of treatment. Given the structure (17), two-way flows are only possible if individuals are endowed with a vector of covariate specific selection unobservables.

Assume that each individual has a single selection unobservable V_i, and consider the following alternative representation of the model 17

$$D_i(x, Z) = \mathbb{1}\{p(x, Z) \geq U_{ix}\} \qquad U_i \sim U(0, 1) \tag{18}$$

where $p(x,z)$ is the propensity score. With binary Z, for any $X=x$ there is a pair of propensity scores $p(x,1)$ and $p(x,0)$, and these propensity scores can change in arbitrary ways as X varies. Figure 4 illustrates. U_i is individual i's selection unobservable. Their potential treatment choices are $(D(x,1)=1, D(x,0)=0)$ and $(D(x',1)=1, D(x',0)=1)$, so they are a complier at x and an always taker at x'.

As U_i stays fixed for all individual's when X changes, there can be no two-way flows. If $\min(p(x,0),p(x,1)) < \min(p(x',0),p(x',1))$, then moving from x to x' will lead some compliers to become always takers, while moving from x' to x will lead to a one-way flow in the other direction. However, there is no change that simultaneously induces some units to move from complier to always-taker and other to move from always-taker to complier.

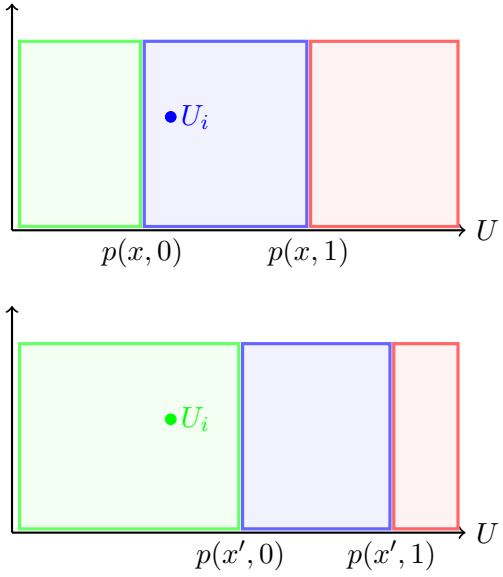


Figure 4: Potential Choices for $X=x$ and $X=x'$ with a single U_i

To see that the V_{ix} may not be independent of X, consider again the selection model (17),

$$\begin{aligned} D(X=0,Z) &= \mathbb{1}\{\alpha_0 + \delta_0 Z \geq V_{0i}\} && X=0 \\ D(X=1,Z) &= \mathbb{1}\{\alpha_1 + \delta_1 Z \geq V_{1i}\} && X=1 \end{aligned}$$

Imbens and Angrist (1994) do not impose that potential treatment choices and outcomes are independent of X. This implies that the vector (V_{0i}, V_{1i}) is not drawn independently of X. The

selection model gives a direct mapping from potential outcomes to thresholds

$$\begin{aligned}
D(0,0) = 0 &\iff V_{0i} > \alpha_0 & D(0,0) = 1 &\iff V_{0i} \leq \alpha_0 \\
D(0,1) = 0 &\iff V_{0i} > \alpha_0 + \delta_1 & D(0,1) = 1 &\iff V_{0i} \leq \alpha_0 + \delta_1 \\
D(1,0) = 0 &\iff V_{1i} > \alpha_1 & D(1,0) = 1 &\iff V_{1i} \leq \alpha_1 \\
D(1,1) = 0 &\iff V_{1i} > \alpha_1 + \delta_1 & D(1,1) = 1 &\iff V_{1i} \leq \alpha_1 + \delta_1
\end{aligned}$$

So, for example,

$$\Pr(D(0,0) = 0, D(0,1) = 1, D(1,0) = 1, D(1,1) = 1) = \Pr(\alpha_0 < V_{0i} \leq \alpha_0 + \delta_0, V_{1i} \leq \max(\alpha_1, \alpha_1 + \delta_1))$$

It follows immediately that if the distribution of (V_1, V_0) is independent of X, then potential outcomes will be too.

To obtain the final model, combining the selection model (17) for $x = 0$ and $x = 1$,

$$D_i(X, Z) = \mathbb{1}\{(1 - X)(\alpha_0 + \delta_0 Z) + X(\alpha_1 + \delta_1 Z) \geq (1 - X)V_{0i} + XV_{1i}\}$$

and rewrite as

$$D_i(X, Z) = \mathbb{1}\{\alpha_0 + \delta_0 Z + \underbrace{\alpha_1 - \alpha_0 + V_{0i} - V_{1i}}_{a_i} X + \underbrace{\delta_1 - \delta_0}_{b} XZ \geq V_{0i}\}$$

C.2 A Selection Model

In Section 2.1 we saw that $s(X)$ is identified by the sign of the difference in propensity scores $p(X, 1) - p(X, 0)$. This property can be exploited to simplify the estimation of $s(X)$.

Recall the equivalence result of Vytlacil (2002) implies that, for each value of X

$$p(X, Z) = F_{V_X|X}(\alpha_X + \delta_X Z)$$

where $F_{V|X}(\cdot)$ is the distribution of the idiosyncratic selection unobservable V_X. Notice that, as $F_{V|X}(\cdot)$ is a monotonic function, for a fixed $X = x$,

$$\text{sign}(p(x, 1) - p(x, 0)) = \text{sign}(\delta_x).$$

With no further assumptions, this simplification is of limited use. Consider $(x, x') \in \mathcal{X}^2$ with $x \neq x'$. Consider $p(x, 0)$ and $p(x', 0)$. The selection model implies that $p(x, 0) = F_{V_x|x}(\alpha_x)$ and $p(x', 0) = F_{V_{x'}|x'}(\alpha_{x'})$. The difference $p(x, 0) - p(x', 0)$ provides no information about α_x and $\alpha_{x'}$ as the distribution function may shift.

Theorem 4 (Selecton Model) *Assumption 8 implies and is implied by a selection model of the form*

$$D(X,Z) = \mathbb{1}\{\alpha_X + Z\delta_X \geq V_X\}, \qquad V_X \perp\!\!\!\perp (X,Z), \tag{19}$$

for all X. Furthermore, the marginal distribution of V_X may be assumed to be the same for all values of X.

A full proof is presented in Appendix C. The main argument is a generalisation of the proof of Vytlacil (2002). Consider a unit i. This unit is endowed with a vector $V_i = (V_{0i}, V_{1i}, V_{2i}, ...)$ that characterises their idiosyncratic preferences over treatment for each value of X, and a vector (x_i, z_i) containing their observed covariates and instrument. It is immediate that the selection model (19) implies Assumption 8. Fixing α_X, δ_X and V_{Xi} gives potential treatment choices for $Z = 0$ and $Z = 1$. As α_X and δ_X are constant and V_{Xi} is drawn independently of X and Z, these potential treatment choices will also be independent of X and Z.

To see that Assumption 8 implies a model of the form (19). Fix $X = x$. As $D(x,z)$ is assigned independently of (X,Z),

$$p(x,z) = E[D|X=x,Z=z] = E[D(x,0)|X=x,Z=z] = E[D(x,z)] = E[D(x,z)|X=x',Z=z']$$

Where (x',z') can be any values of X and Z. Choose an arbitrary distribution $F_{V_x}(\cdot)$ for V_X. Given $F_{V_x}(\cdot)$, $\alpha_x = F^{-1}_{V_x}(p(x,0))$ and $\delta_x = F^{-1}_{V_x}(p(x,1)) - \alpha_x$. Note that, under the weaker random assignment condition of Assumption 1, $E[D(x,z)|X=x',Z=z']$ would vary with (x',z'). In place of α_x and δ_x, for each combination of x and x' there would be a pair of parameters $(\alpha_{x,x'}, \delta_{x,x'})$

$$\alpha_{x,x'} = F^{-1}_{V_x}(E[D(x,0)|X=x',Z=0]),$$
$$\delta_{x,x'} = F^{-1}_{V_x}(E[D(x,1)|X=x',Z=1]) - \alpha_{x,x'}.$$

Furthermore, only $\alpha_{x,x}$ and $\delta_{x,x}$ would be identified by the data.

We can then proceed to construct V_{xi} in the same manner as Vytlacil (2002). Assume that switchers for x are compliers, in which case $p(x,1) > p(x,0)$ and $\delta_x > 0$.

- If $D_i(x,0) = D_i(x,1)$ then i is an always taker when $X_i = x$. Draw U_{ix} from the uniform distribution with bounds $(0, p(x,0))$. Note that
- If $D_i(x,0) = 0$ and $D_i(x,1) = 1$, then i is a complier when $X_i = x$. Draw U_{ix} from the uniform distribution with bounds $(p(x,0), p(x,1))$. Set $V_{xi} = F^{-1}_{V_x}(U_{xi})$.
- If $D_i(x,0) = D_i(x,1) = 0$, then i is a never taker when $X_i = x$. Draw U_{ix} from the uniform distribution with bounds $(p(x,1), 1)$.

Set $V_{xi} = F_{V_x}^{-1}(U_{xi})$. In this construction, U_{xi} is determined by the potential outcomes $(D(x,0), D(x,1))$, which are independent of (X, Z), and two constants $p(x,0)$ and $p(x,1)$. Hence, U_{xi} is independent of (X, Z). Since $F_{V_x}(\cdot)$ is chosen arbitrarily, it can be chosen to be the same for all values of X. This process can be repeated for all values of X, and the result follows.

Figure 5 shows an example. There are two covariate values x and x'. Switchers are compliers at x, which means that $p(x,0) < p(x,1)$. If $X_i = x$, unit i would be a complier, so U_{xi} is between $p(x,0)$ and $p(x,1)$. Switchers at x' are defiers, so $p(x',0) > p(x',1)$. In this case, unit i would be a never taker, and $U_{x'i}$ is above $p(x',0)$.

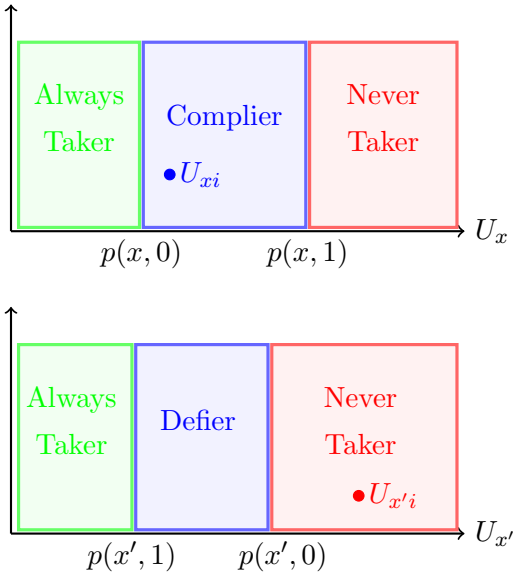


Figure 5: U_{xi} and $U_{x'i}$ when unit i is a complier if $X_i = x$ and a never taker if $X_i = x'$.

The representation (19) exists even if, in the true data generating process, the distributions of V_x and $V_{x'}$ differ. Intuitively, we can only estimate the propensity score, $p(x,z) = F_{V_x}(\alpha_x + \delta_x z)$, and this does not separately identify the parameters α_x and δ_x, and the distribution $F_{V_x}(\cdot)$. Hence, we can always fix the distribution to be common, and adjust the parameters appropriately. Say that the true process generating potential treatment choices for $X = x'$ is

$$D(x', Z) = \mathbb{1}\{a_{x'} + d_{x'} Z \geq V_{x'}\},$$

with $F_{V_{x'}}(\cdot)$ different from $F_{V_x}(\cdot)$. Set $\alpha_{x'} = F_{V_x}^{-1}(F_{V_{x'}}(a_{x'}))$ and $\delta_{x'} = F_{V_x}^{-1}(F_{V_{x'}}(a_{x'} + d_{x'})) - \alpha_{x'}$, then

$$\begin{aligned}
p(x,0) &= F_{V_{x'}}(a_{x'}) = F_{V_x}(F_{V_x}^{-1}(F_{V_{x'}}(a_{x'}))) = F_{V_x}(\alpha_{x'}), \\
p(x,1) &= F_{V_{x'}}(a_{x'} + d_{x'}) = F_{V_x}(F_{V_x}^{-1}(F_{V_{x'}}(a_{x'} + d_{x'}))) = F_{V_x}(\alpha_{x'} + \delta_{x'}).
\end{aligned}$$

Hence, a process where $V_{x'}$ is generated according to $F_{V_x}(\cdot)$ and the parameters of the selection model are $\alpha_{x'}$ and $\delta_{x'}$ would be observationally equivalent to the true data generating process.